



Submersible Borehole Pump

UPA S

Size: UPA S 350

Nominal Pump Diameter: 14 Inches

Type Series Booklet



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Type Series Booklet UPA S

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Water Supply

Submersible Borehole Pumps

UPA S 350



Main applications

- Spray irrigation systems
- General irrigation systems
- Pressure boosting
- Groundwater management
- Water supply systems
- Offshore processes

Fluids handled

- Drinking water
- River water, lake water and groundwater
- Maximum sand content in fluid handled depending on particle size and duration, sand content up to 250 g/m³¹⁾

Operating data

Table 1: Operating properties

Characteristic		Value		
		50 Hz	60 Hz	100 Hz
Flow rate	Q [m ³ /h]	≤ 935	≤ 874	≤ 967
	Q [l/s]	≤ 260	≤ 243	≤ 269
Head	H [m]	≤ 358	≤ 373	≤ 384
Fluid temperature	T [°C]	≤ 50	≤ 50	≤ 50
Speed	n [rpm]	≤ 2900	≤ 3500	≤ 3000
Well diameter	D [mm]	≥ 350	≥ 350	≥ 350
	D ["]	≥ 14	≥ 14	≥ 14

Design details

Design

- Centrifugal pump
- Single-stage or multistage
- Single-suction
- Ring-section design
- Rigid connection between pump and motor

Type of installation

- Vertical installation
- Horizontal installation
- Angled installation

Drive

Asynchronous motor:

- With squirrel cage motor for submerged use
- Standard connection to NEMA
- Frequency 50 Hz/60 Hz
- IP68 enclosure
- DOL starting or star-delta starting
- Frequency of starts ≤ 15 starts per hour
- Winding J1 (PPC/HDPE) or, for higher temperatures, J2 (radiation-crosslinked PE2/PA)

Synchronous motor:

- Interior (buried) permanent magnet synchronous motor (IPMSM)
- IP68 enclosure
- Standard connection to NEMA
- Frequency 100 Hz
- Winding J2 (radiation-crosslinked PE2/PA)
- Frequency of starts ≤ 15 starts per hour

Electrical connection

- Delivered ex-factory with 1 or 2 motor leads (including earth conductor and internal earthing)
- Extension cable connected with water-tight cable connector
- Motor lead and extension cable suitable for drinking water use

Impeller type

- Mixed flow design

Bearings

- Radial plain bearings
- Pump bearings lubricated by fluid handled; motor bearings lubricated by water fill
- Axial thrust is balanced by a tilting-pad thrust bearing in the motor (lower end)

¹ For 50 g/m³ and above, the more wear-resistant version with silicon carbide bearings is recommended.

Connections

- Pump screw-ended or flanged
- Threaded end to DIN ISO 228, Part 1 (up to 6 inch)
- NPT threaded end to ASME B1.20.1-2013
- Flanged end to DIN EN 1092
- With lift check valve or connection nozzle

Designation

Table 2: Designation example

Position																																
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	
U	P	A	S		3	5	0	-	5	0	0	/	0	1	C	C	+	U	M	A	S	2	0	0	-	7	5	/	4	2	C	
Pump																	+	Motor														
Pump set																																

Table 3: Designation key

Position	Code	Description
1-5	Type series	
	UPA S	Precision (investment) cast variant
6-8	Nominal pump diameter [mm]	
	350	
9-12	Flow rate at best efficiency point [m³/h]	
	-500	500
	-650	650
13	/	
14-15	Number of stages	
	01, 02, 03, 04, 05, 06, 07	
16	Pump material	
	D	Duplex 1.4517
	C	Stainless steel AISI 316
17	Impeller material	
	D	Duplex 1.4517
	C	Stainless steel AISI 316
18	+	With motor
	-	Without motor
19-22	Motor version	
	UMA	Asynchronous motor
	UMA-S	Permanent magnet synchronous motor
23-25	Nominal motor diameter [mm]	
	200, 250, 300	
26	-	
27-28	Motor rating PN [kW]	
	37	37
	...	...
	400	400
29	/	
30	Number of motor poles	
	2	2 poles
	4	4 poles
31	Winding insulation	
	1	J1 (PPC/HDPE)
	2	J2 (radiation-crosslinked PE2/PA)
32	Motor material	
	D	1.4539 / AISI 904L
	C	Stainless steel AISI 316
	G	Grey cast iron

Materials

Table 4: Pump material selection, UPA S 350

Component	Material variant	
	CC	DD
Impeller	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Suction casing	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Screws, bolts and nuts	CrNiMo steel (A4)	CrNiMo steel (1.4462)
Stage casing	CrNiMo steel (1.4408)	CrNiMo steel (1.4517)
Shaft	CrNiMo steel (1.4462)	CrNiMo steel (1.4462)

Table 5: Motor material selection, UMA 200, UMA-S 200, UMA 250, UMA-S 250, UMA 300

Component	Size	Material variant		
		G	C	D
Casing	UMA 200, UMA-S 200	Grey cast iron (EN-GJL-200)	CrNiMo steel (1.4408)	CrNiMo steel (1.4539)
	UMA 250, UMA-S 250	Grey cast iron (EN-GJL-200)	CrNiMo steel (1.4408)	CrNiMo steel (1.4539)
	UMA 300	Grey cast iron EN-GJL-250	-	CrNiMo steel (1.4517)
Screws, bolts and nuts	UMA 200, UMA-S 200	CrNiMo steel (A4-70)	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 250, UMA-S 250	CrNiMo steel (A4-70)	CrNiMo steel (A4-70)	CrNiMo steel (1.4539)
	UMA 300	CrNiMo steel (1.4571)	-	CrNiMo steel (1.4462)
Shaft	UMA 200, UMA-S 200	CrNiMo steel (1.4460)	CrNiMo steel (1.4460)	CrNiMo steel (1.4462)
	UMA 250, UMA-S 250	CrNiMo steel (1.4460)	CrNiMo steel (1.4460)	CrNiMo steel (1.4462)
	UMA 300	CrNiMo steel (1.4462-C45+N)	-	CrNiMo steel (1.4462-C45+N)

Product benefits

- High efficiencies through optimised, precision (investment) cast hydraulic design
- High operating reliability as all components are made of high-quality stainless steel.
- Overall efficiency can be increased further by combining the pump with an UMA S synchronous motor with ratings of up to 250 kW
- Reliable operation by anti-jam lift check valve
- Easy to install with service-friendly hydraulic system design
- Solids-induced wear can be minimised by metal wear rings and bearings with enhanced wear resistance (optional).

Certifications

Table 6: Overview

Label	Effective in:	Comment
	All countries	Certified quality management to ISO 9001
	France	Approved in accordance with the French drinking water regulation

Product information as per Regulation No. 1907/2006 (REACH)

For information as per European chemicals regulation (EC) No. 1907/2006 (REACH) see <https://www.ksb.com/en-global/company/corporate-responsibility/reach>.

Overview of submersible motors

The following submersible motors are available:

Table 7: Selection table

Well diameter	Motor size	Voltage range	Rated power
8 inches	UMA 200	Up to 1 kV	37 to 90 kW
8 inches	UMA-S 200	400 V	45 to 130 kW
10 inches	UMA 250	Up to 1 kV	85 to 190 kW
10 inches	UMA-S 250	400 V	150 to 250 kW
12 inches	UMA 300	Up to 1 kV	250 to 400 kW

- Further motor data for the UMA asynchronous motor see type series booklet No. 3455.51
- Further motor data for the UMA-S synchronous motor see type series booklet No. 3455.52

Technical data, 50 Hz

Information on characteristic curves

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Selection chart UPA S 350, 50 Hz, n = 2900 rpm

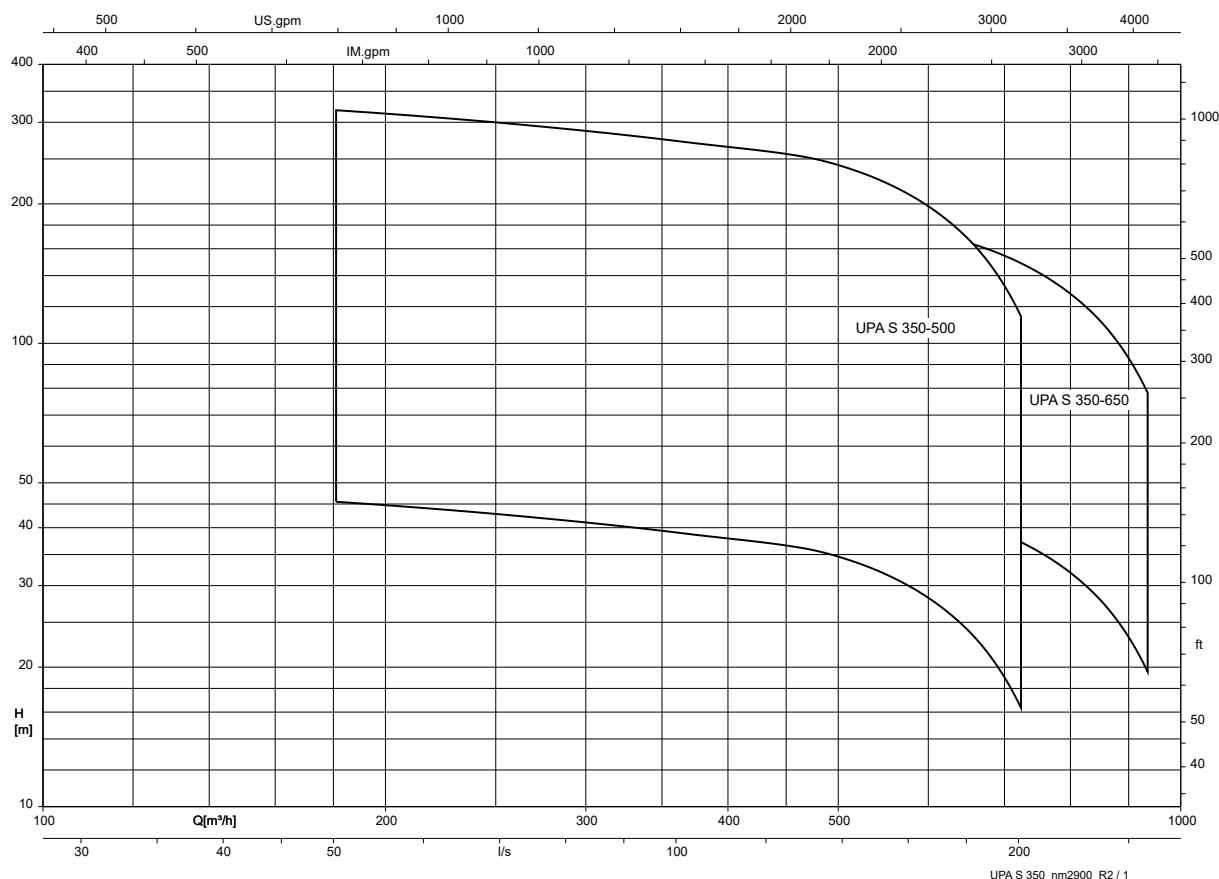
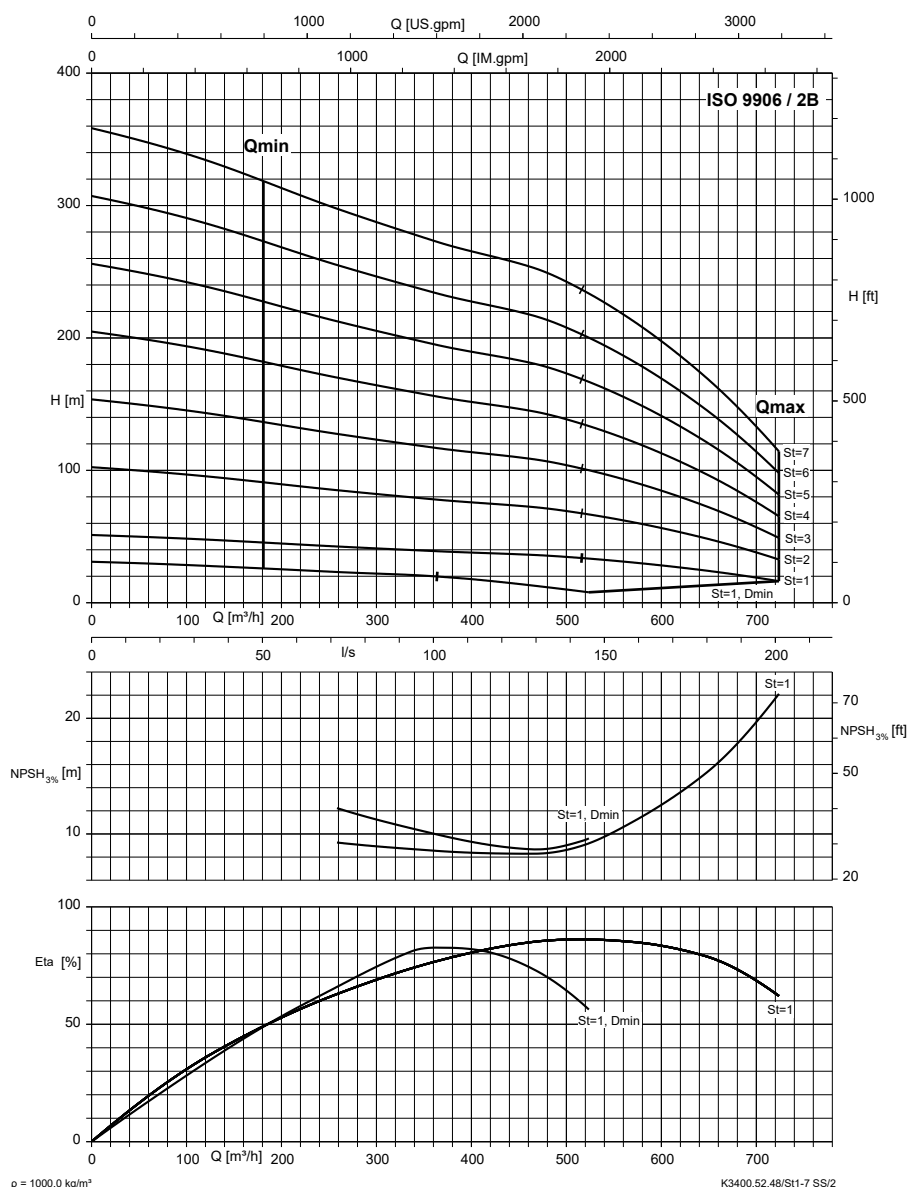


Fig. 1: Selection chart UPA S 350, 50 Hz

UPA S 350 - 500 / ..., number of stages 1 - 7, 50 Hz, n = 2900 rpm



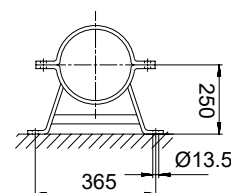
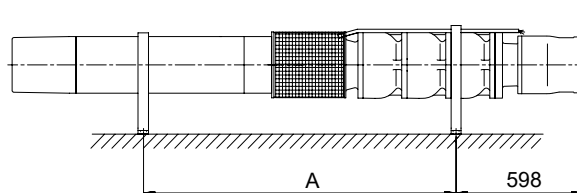
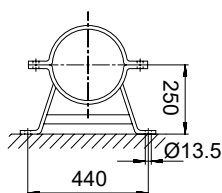
Operating range

$$Q_{\min} = 181 \text{ m}^3/\text{h}$$

$Q_{\min}$ = 10 m³/h
 $Q_{\max}$ = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4) (⇒ Fig. 5)



Dimensions of UPA S 350 - 500 / ... [mm]

Connection types

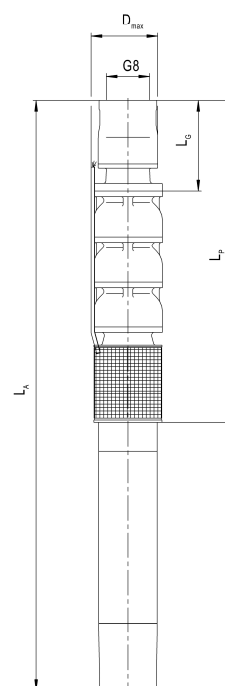
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

 On versions with an alternative pump end, dimensions $D_{\max}$ and L_A differ from those with the standard pump end.

For alternative pump end dimensions see:
(⇒ Table 23)



3400.515/01-EN

Table 8: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 350 - 500 / ...	A	L _P ²⁾	L _A ²⁾	L _G ³⁾²⁾	D _{max} ²⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 200-65/21	1125	988	2458	395	306	306	337	X	X
2 + UMA 250-132/21	1443	1211	2870	395	308	308	541	X	X
3 + UMA 250-190/21	-	1434	3353	395	325	325	668	X	-
4 + UMA 300-250/22	2106	1667	3740	395	336	336	840	X	X
5 + UMA 300-300/22	-	1890	4143	395	336	336	958	X	-
6 + UMA 300-400/22	-	2113	4486	395	336	336	1049	X	-
7 + UMA 300-400/22	-	2336	4709	395	336	336	1089	X	-

Table 9: Technical data, 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

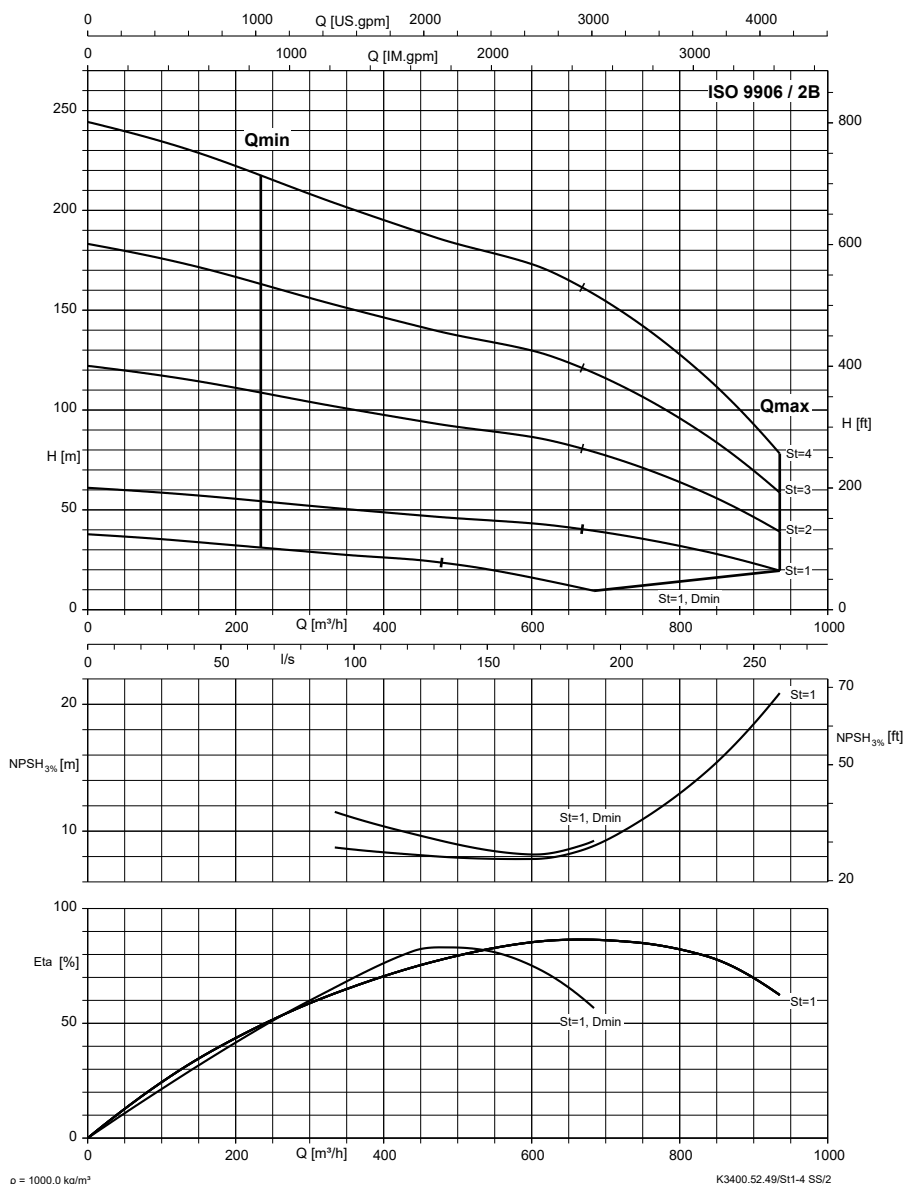
UPA S 350 - 500 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ⁴⁾	
						DOL	Y - Δ
						[mm²]	[mm²]
1 + UMA 200-65/21	51	56	65	128	87,7	F3/F4 × 10 II	F3/F4 × 10
2 + UMA 250-132/21	102	111	132	249	89,0	F3/F4 × 25 II	F3/F4 × 25
3 + UMA 250-190/21	154	167	190	365	89,7	F3/R4 × 35 II	F3/R4 × 35
4 + UMA 300-250/22	205	222	250	465	88,4	F3/F3 × 70 II + R1 × 95	6 × R1 × 95 + R1 × 95
5 + UMA 300-300/22	256	278	300	558	89,2	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
6 + UMA 300-400/22	307	333	400	739	89,6	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
7 + UMA 300-400/22	358	389	400	739	89,6	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 10: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
R1	1 × 1-core, cable cross-section: round
F3/F3	1 × 3-core, cable cross-section: flat + 1 × 3-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

- 2 Dimension indicated refers to version with lift check valve and standard pump end G8.
3 Length of pump end fitted as standard
4 Designed for submerged use, 400 V, ≤ +30 °C

UPA S 350 - 650 / ..., number of stages 1 - 4, 50 Hz, n = 2900 rpm



Connection types

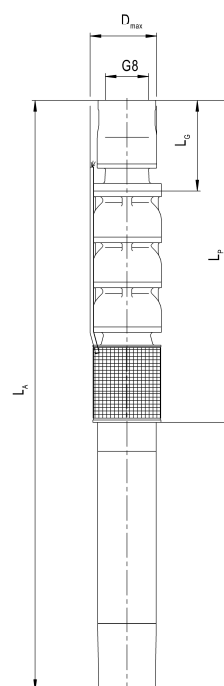
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 24)



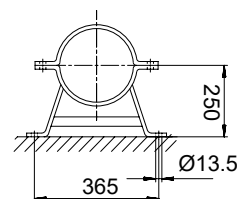
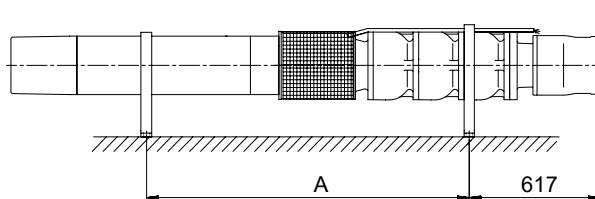
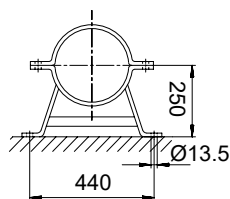
Operating range

$Q_{min} = 234 \text{ m}^3/\text{h}$

$Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 6) (⇒ Fig. 7)



Dimensions of UPA S 350 - 650 / ... [mm]

3400.515/01-EN

Table 11: Dimensions, weights and installation type depending on the motor, 50 Hz / 400 V 3~

UPA S 350 - 650 / ...	A	L _P ⁵⁾	L _A ⁵⁾	L _G ⁶⁾⁵⁾	D _{max} ⁵⁾		Total weight	Type of installation	
					DOL	Y - Δ			
					[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]	Vertical	Horizontal
1 + UMA 200-90/21	-	1007	2747	395	322	322	394	✗	-
1 + UMA 250-110/21	1155	1007	2536	395	342	342	474	✗	✗
2 + UMA 250-190/21	-	1249	3168	395	342	342	646	✗	-
3 + UMA 300-300/22	2024	1501	3754	395	358	358	904	✗	✗
4 + UMA 300-400/22	-	1743	4116	395	358	358	1003	✗	-

Table 12: Technical data, 50 Hz / 400 V 3~, n = 2900 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 350 - 650 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ⁷⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA 200-90/21	61	86	90	180	88,0	F3/F4 × 16 II	F3/F4 × 16
1 + UMA 250-110/21	61	86	110	215	88,5	F3/F4 × 16 II	F3/F4 × 25
2 + UMA 250-190/21	122	171	190	215	88,5	F3/R4 × 35 II	F3/R4 × 35
3 + UMA 300-300/22	183	257	300	558	89,2	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
4 + UMA 300-400/22	244	342	400	739	89,6	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 13: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
R1	1 × 1-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

⁵⁾ Dimension indicated refers to version with lift check valve and standard pump end G8.

⁶⁾ Length of pump end fitted as standard

⁷⁾ Designed for submerged use, 400 V, ≤ +30 °C

Technical data, 60 Hz

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Information on characteristic curves

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Selection chart UPA S 350, 60 Hz, n = 3500 rpm

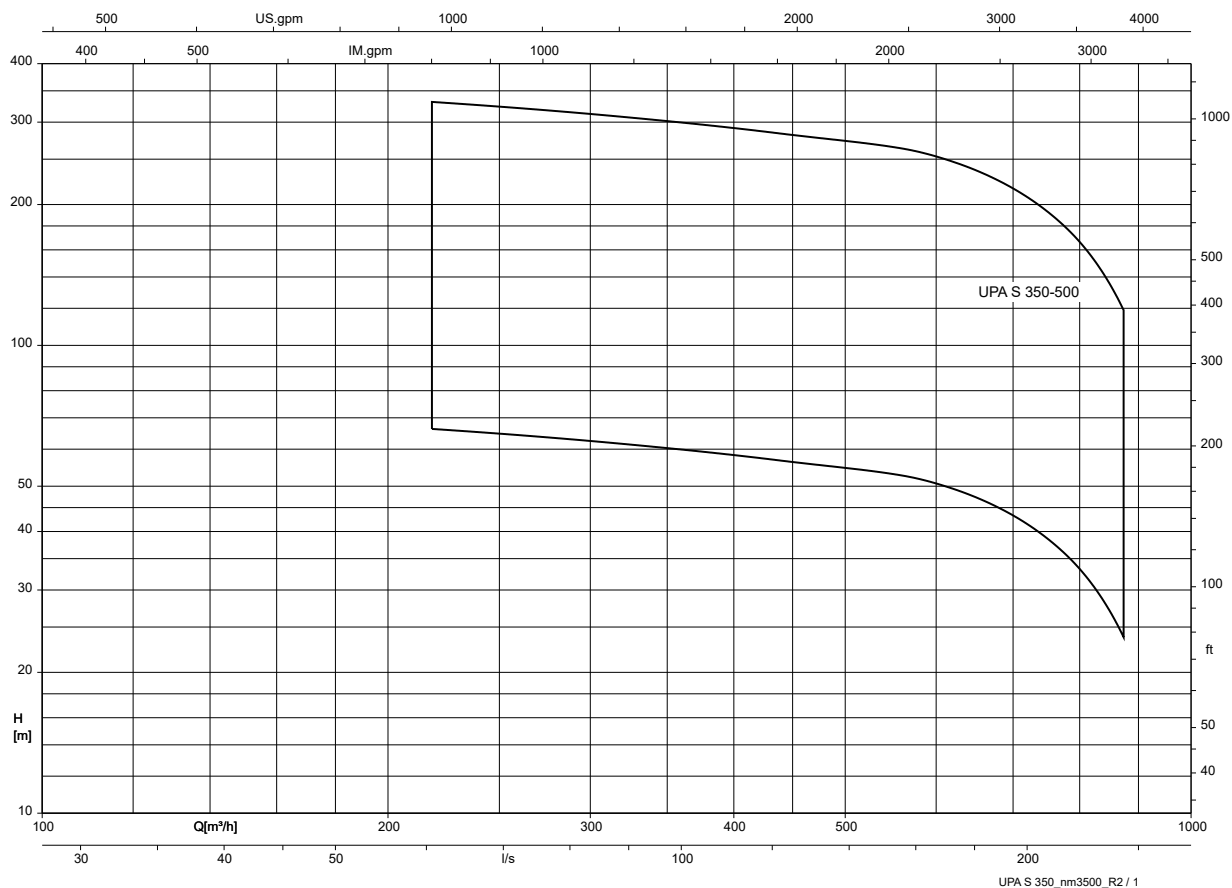
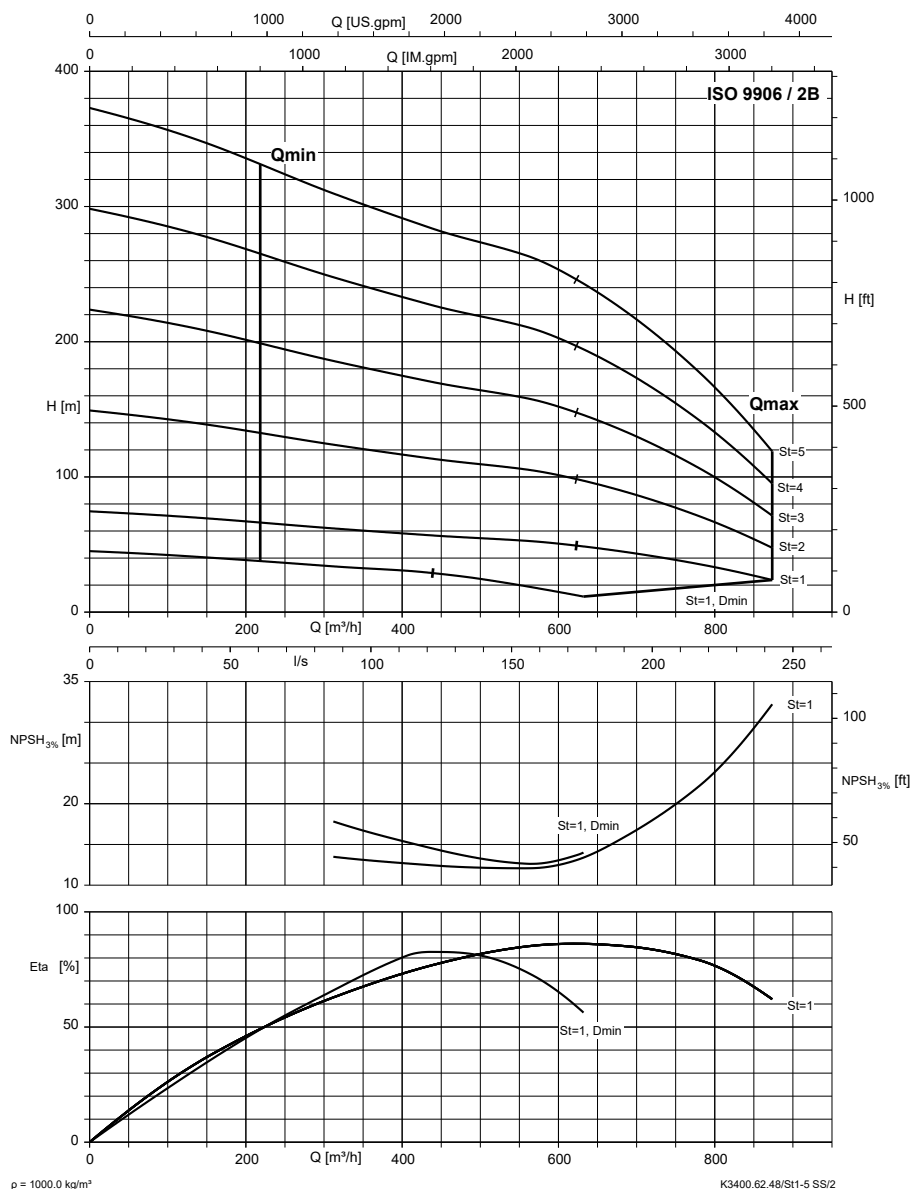


Fig. 2: Selection chart UPA S 350, 60 Hz

UPA S 350 - 500 / ..., number of stages 1 - 5, 60 Hz, n = 3500 rpm



Operating range

$Q_{min} = 218 \text{ m}^3/\text{h}$

$Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4) (⇒ Fig. 5)

Connection types

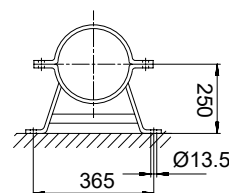
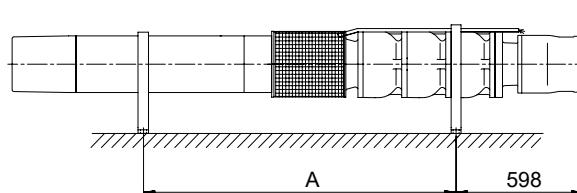
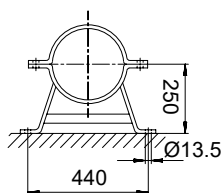
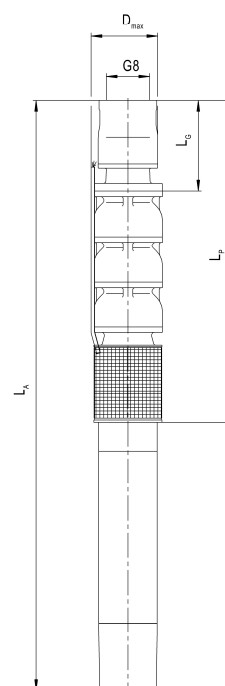
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 23)



Dimensions of UPA S 350 - 500 / ... [mm]

Table 14: Dimensions, weights and installation type depending on the motor, 60 Hz / 460 V 3~

UPA S 350 - 500 / ...	A	L _P ⁸⁾	L _A ⁸⁾	L _G ⁹⁾⁸⁾	D _{max} ⁸⁾		Total weight	Type of installation	
					DOL	Y - Δ		Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	Material variant 1.4408 / 1.4517 [kg]		
1 + UMA 200-90/21	-	988	2728	395	305	305	385	X	-
1 + UMA 250-110/21	1155	988	2517	395	305	308	457	X	X
2 + UMA 250-190/21	-	1211	3130	395	325	325	629	X	-
3 + UMA 300-300/22	1973	1444	3697	395	336	336	878	X	X
4 + UMA 300-400/22	-	1667	4040	395	336	336	970	X	-
5a* + UMA 300-400/22 ¹⁰⁾	-	1890	4263	395	336	336	1010	X	-

Table 15: Technical data, 60 Hz / 460 V 3~, n = 3500 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 350 - 500 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹¹⁾	
	H ₀	P ₂	P _N	I _N	η _M	DOL	Y - Δ
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA 200-90/21	75	97	108	179	88,2	F3/F4 × 16 II	F3/F4 × 16
1 + UMA 250-110/21	75	97	132	219	88,7	F3/F4 × 16 II	F3/F4 × 25
2 + UMA 250-190/21	149	193	228	371	90,0	F3/R4 × 35 II	F3/R4 × 50
3 + UMA 300-300/22	224	290	360	585	89,3	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
4 + UMA 300-400/22	298	387	420	682	89,9	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95
5a* + UMA 300-400/22 ¹⁰⁾	325	391	420	682	89,9	6 × R1 × 95 II + R1 × 95	6 × R1 × 95 + R1 × 95

Table 16: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
R1	1 × 1-core, cable cross-section: round
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

8 Dimension indicated refers to version with lift check valve and standard pump end G8.

9 Length of pump end fitted as standard

10 a* = Reduced impeller diameter

11 Designed for submerged use, 460 V, ≤ +30 °C

Technical data, 100 Hz

Tolerance to ISO 9906 Cl. 2B

The characteristic curves shown are valid for:

- Water at 20 °C
- Density 998 kg/m³
- Viscosity 1.00 mm²/s

Information on characteristic curves

The characteristic curves shown refer to pump sets driven by a UMA-S synchronous motor, n = 3000 rpm.

The characteristic curves shown are intended to allow pre-selection. Refer to the quotation for the precise selection data.

Selection chart UPA S 350, 100 Hz, n = 3000 rpm

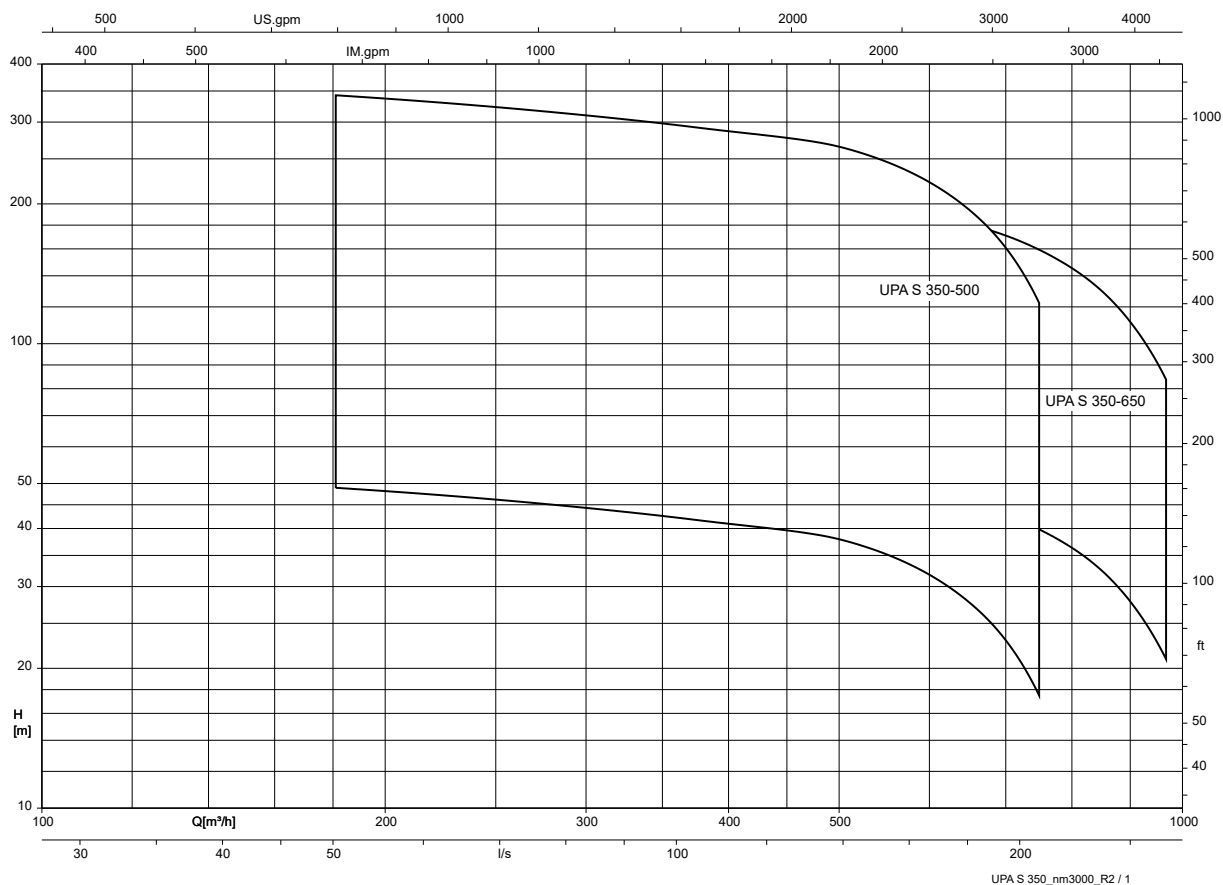
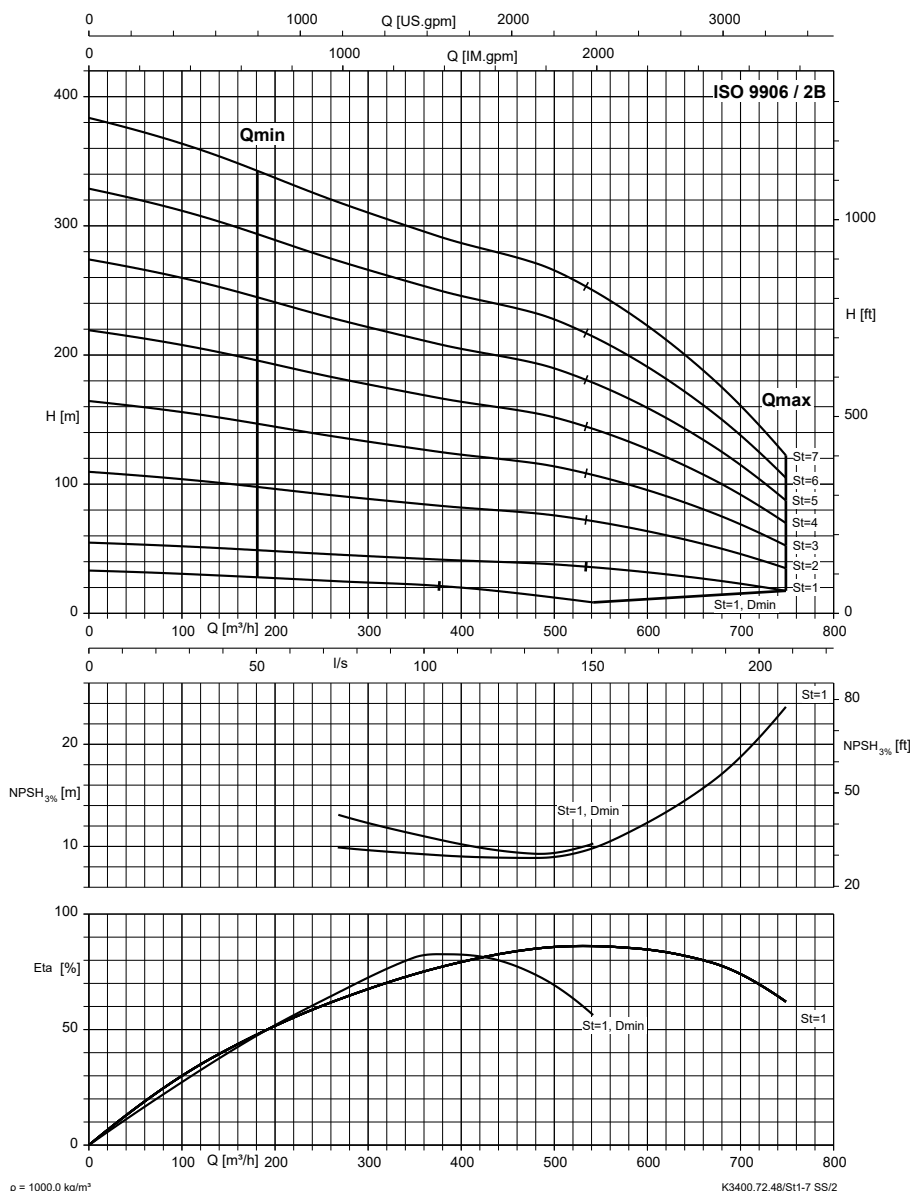


Fig. 3: Selection chart UPA S 350, 100 Hz

UPA S 350 - 500 / ..., number of stages 1 - 7, 100 Hz, n = 3000 rpm



Connection types

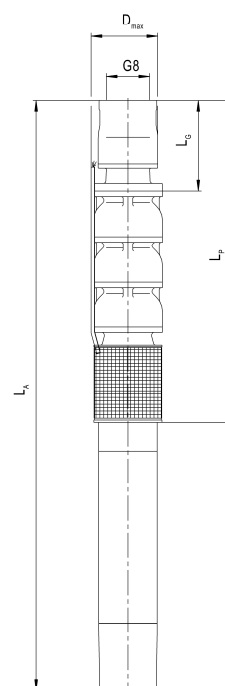
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 23)



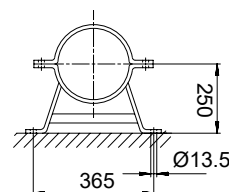
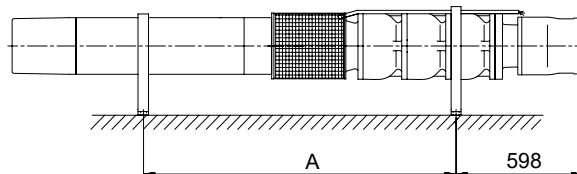
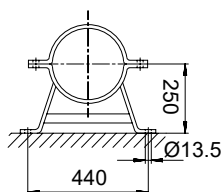
Operating range

$Q_{min} = 181 \text{ m}^3/\text{h}$

Q_{max} = end of stage curve

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 4) (⇒ Fig. 5)



Dimensions of UPA S 350 - 500 / ... [mm]

3400.515/01-EN

Table 17: Dimensions, weights and installation type depending on the motor, 100 Hz, 400 V 3~

UPA S 350 - 500 / ...	A	L _p ¹²⁾	L _A ¹²⁾	L _G ¹³⁾¹²⁾	D _{max} ¹²⁾		Total weight	Type of installation	
					VFD	VFD parallel	Material variant 1.4408 / 1.4517	Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]		
1 + UMA-S 200-75/42	993	988	2193	395	302	305	288	x	x
2 + UMA-S 250-200/42	1188	1211	2980	395	-	325	476	x	x
3 + UMA-S 250-200/42	1666	1434	3203	395	-	325	516	x	x

Table 18: Technical data, 100 Hz, 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 350 - 500 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹⁴⁾	
	H ₀	P ₂	P _N	I _N	η _M	VFD	VFD parallel
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA-S 200-75/42	55	62	75	128	92,8	F4 × 16	F3/F4 × 16 II
2 + UMA-S 250-200/42	110	123	200	389	93,8	-	F3/R4 × 35 II
3 + UMA-S 250-200/42	164	185	200	389	93,8	-	F3/R4 × 35 II

Table 19: Explanation of abbreviations

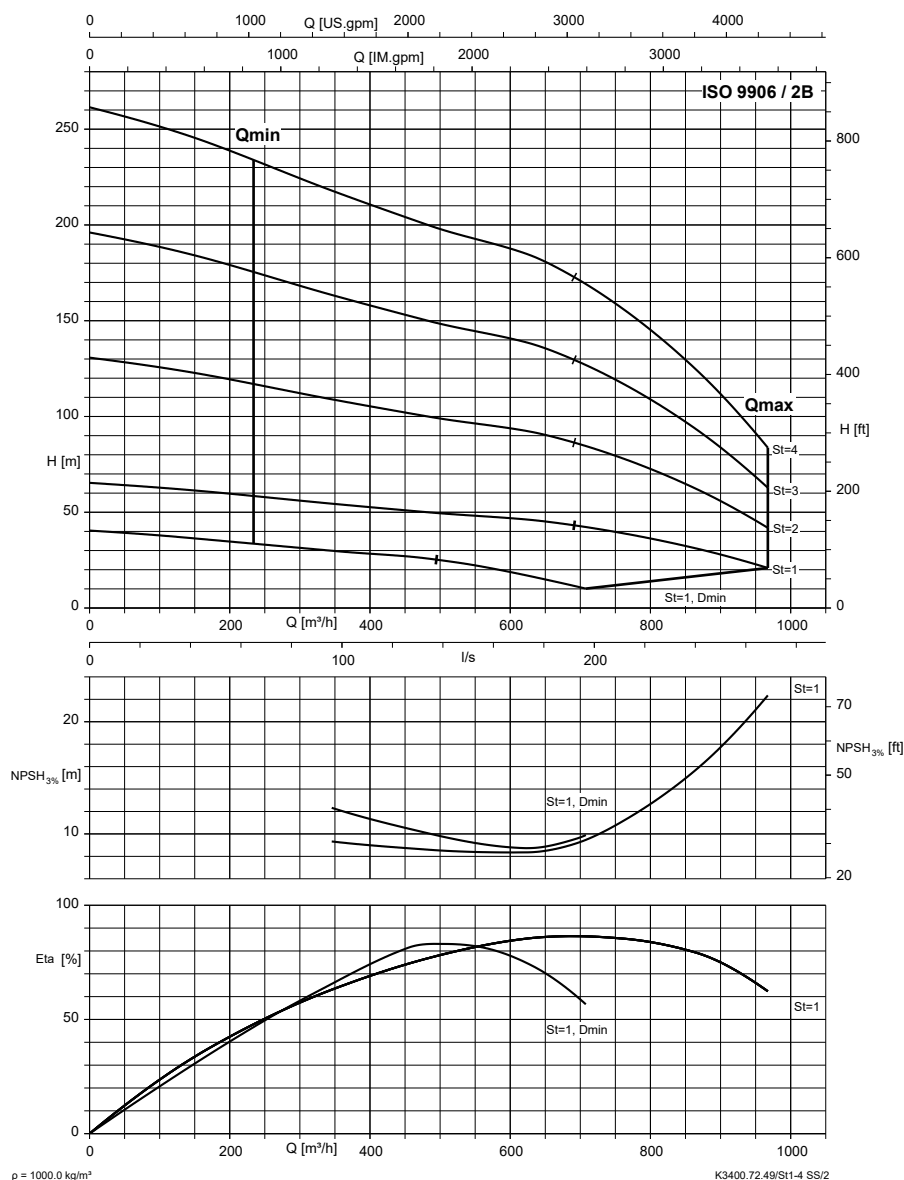
Abbreviation	Explanation
II	Parallel cables
F4	1 × 4-core, cable cross-section: flat
F3/F4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

12 Dimension indicated refers to version with lift check valve and standard pump end G8.

13 Length of pump end fitted as standard

14 Designed for submerged use, 400 V, ≤ +30 °C

UPA S 350 - 650 / ..., number of stages 1 - 4, 100 Hz, n = 3000 rpm



Connection types

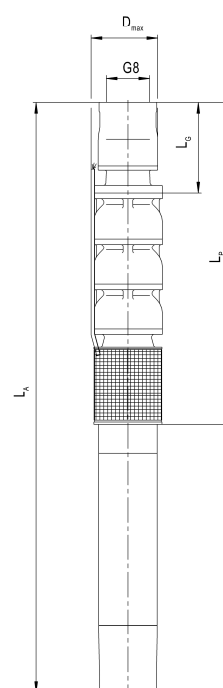
Standard pump end = G8

Alternative pump ends:

- G10
- DN 200, DN 250
- NPT8-8, NPT10-8

i On versions with an alternative pump end, dimensions D_{max} and L_A differ from those with the standard pump end.

For alternative pump end dimensions see: (⇒ Table 24)



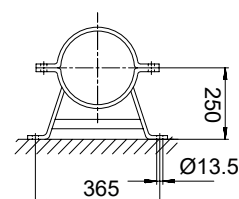
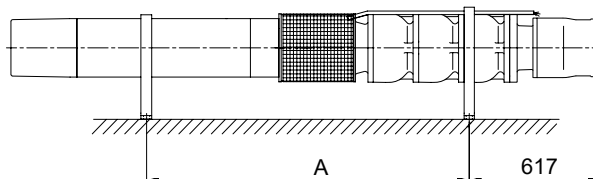
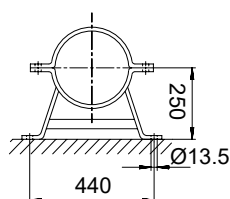
Operating range

$Q_{min} = 234 \text{ m}^3/\text{h}$

$Q_{max} = \text{end of stage curve}$

The characteristic curves do not take into account the head loss caused by the lift check valve.

1. Refer to the corresponding loss curve to determine and take into account the head loss caused by the lift check valve. (⇒ Fig. 6) (⇒ Fig. 7)



Dimensions of UPA S 350 - 650 / ... [mm]

3400.515/01-EN

Table 20: Dimensions, weights and installation type depending on the motor, 100 Hz / 400 V 3~

UPA S 350 - 650 / ...	A	L _p ¹⁵⁾	L _A ¹⁵⁾	L _G ^{16/15)}	D _{max} ¹⁵⁾		Total weight	Type of installation	
					VFD	VFD parallel	Material variant 1.4408 / 1.4517	Vertical	Horizontal
	[mm]	[mm]	[mm]	[mm]	[mm]	[mm]	[kg]		
1 + UMA-S 200-130/42	-	1007	2491	395	336	322	344	x	-
2 + UMA-S 250-250/42	-	1249	3018	395	-	343	547	x	-

Table 21: Technical data, 100 Hz / 400 V 3~, n = 3000 rpm, flow velocity past the motor v = 0.2 m/s

UPA S 350 - 650 / ...	Pump		Motor			Motor lead	
	Head Q = 0 m³/h	Maximum power input	Rated power	Rated current	Efficiency	Number × cross-section of cores ¹⁷⁾	
	H ₀	P ₂	P _N	I _N	η _M	VFD	VFD parallel
	[m]	[kW]	[kW]	[A]	[%]	[mm²]	[mm²]
1 + UMA-S 200-130/42	65	95	130	224	93,3	-	F3/R4 × 35 II
2 + UMA-S 250-250/42	131	190	250	493	93,8	-	6 × R1 × 35 II + R1 × 35

Table 22: Explanation of abbreviations

Abbreviation	Explanation
II	Parallel cables
R1	1 × 1-core, cable cross-section: round
F3/R4	1 × 3-core, cable cross-section: flat + 1 × 4-core, cable cross-section: flat, offset by 90°

15 Dimension indicated refers to version with lift check valve and standard pump end G8.

16 Length of pump end fitted as standard

17 Designed for submerged use, 400 V, ≤ +30 °C

Loss curves for lift check valves

Head losses caused by lift check valve for UPA S 350-500, pump end G8/NPT 8-8/DN 200

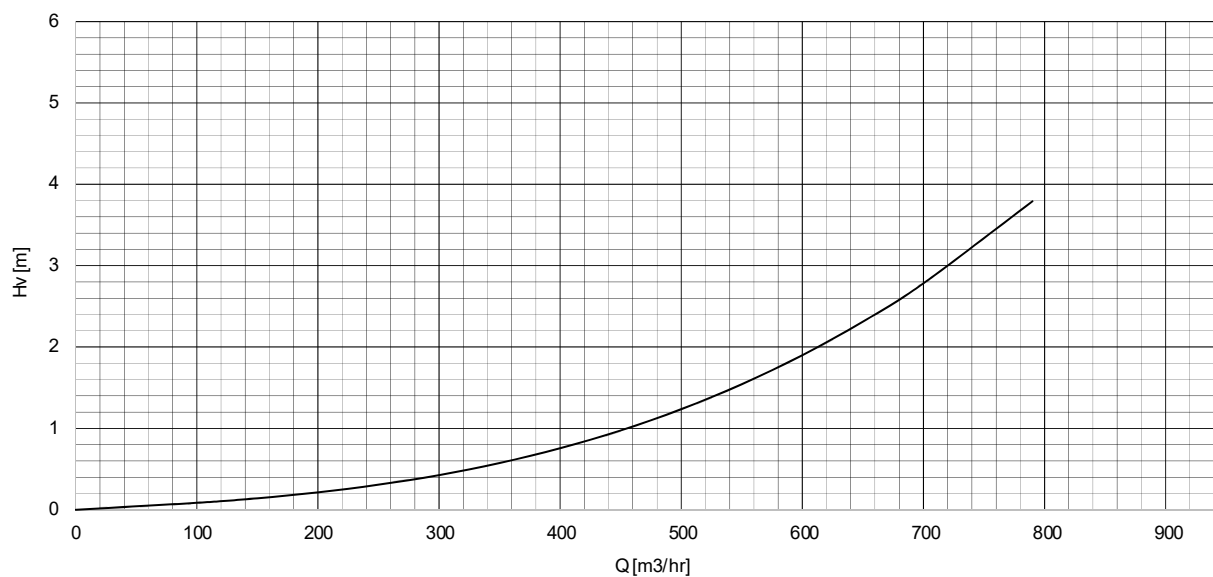


Fig. 4: Loss curve, UPA S 350-500, pump end G8

Head losses caused by lift check valve for UPA S 350-500, pump end G10/NPT 10-8/DN 250

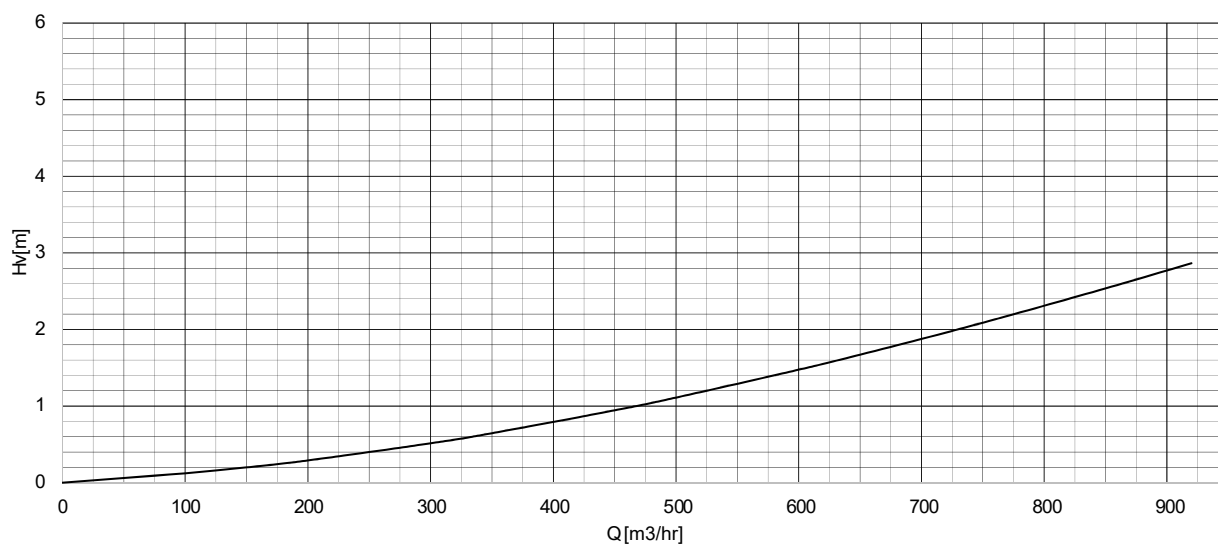


Fig. 5: Loss curve, UPA S 350-500, pump end G10

Head losses caused by lift check valve for UPA S 350-650, pump end G8/NPT 8-8/DN 200

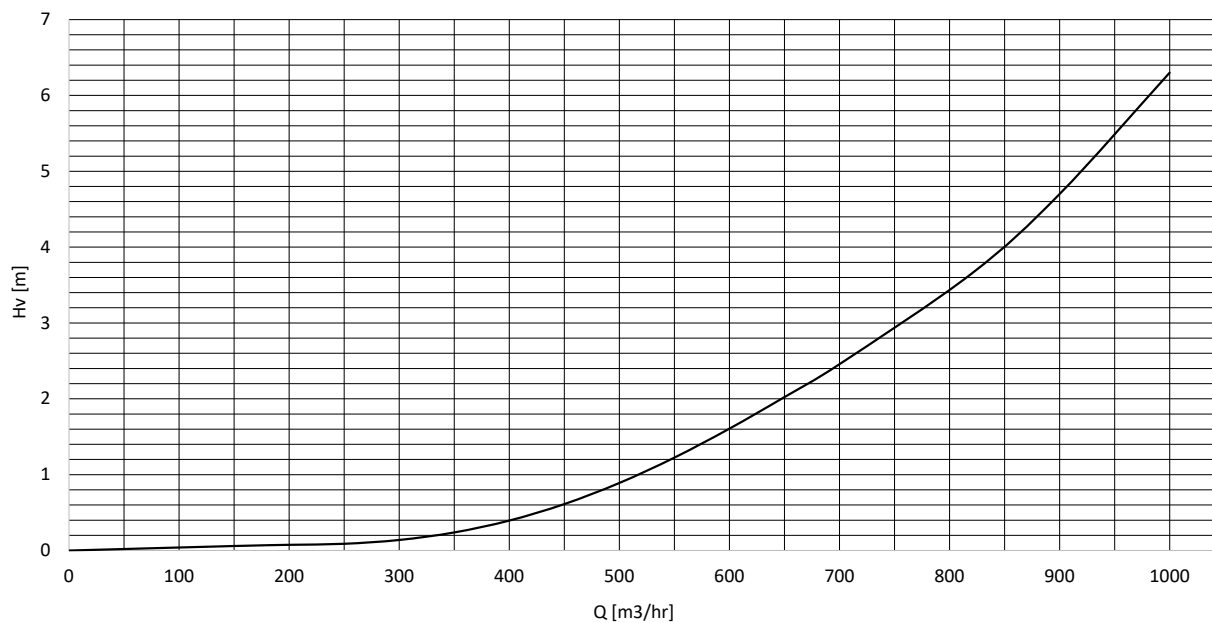


Fig. 6: Loss curve, UPA S 350-650, pump end G8

Head losses caused by lift check valve for UPA S 350-650, pump end G10/NPT 10-8/DN 250

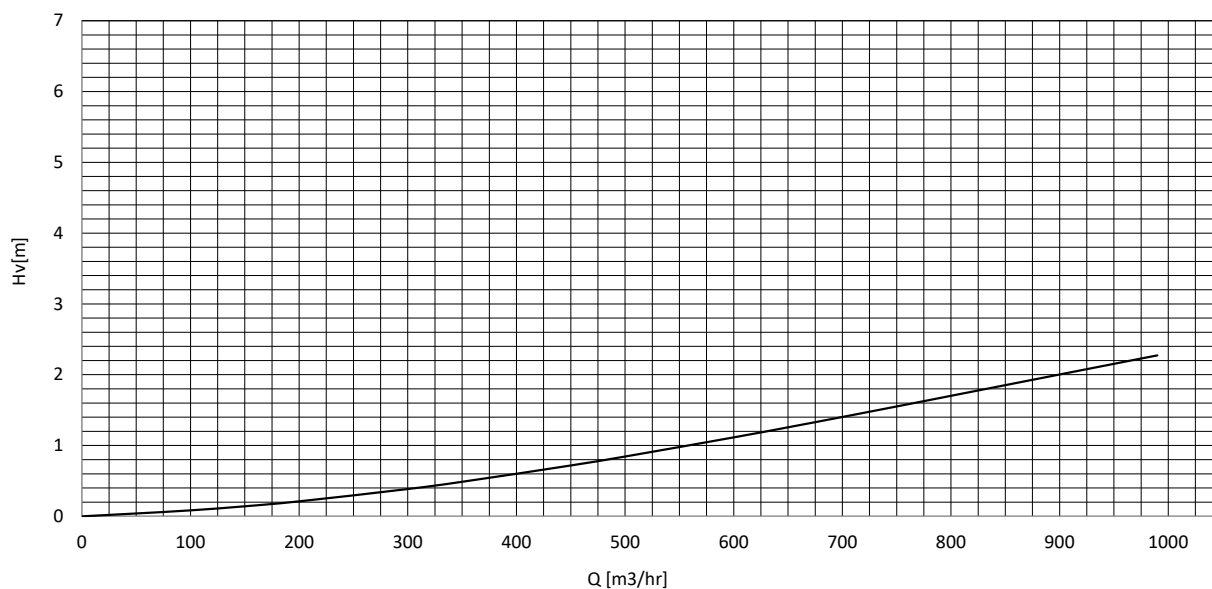


Fig. 7: Loss curve, UPA S 350-650, pump end G10

Dimensions of alternative pump end

i On versions with an alternative pump end, dimensions $D_{\max}^*$ and L_A^* differ from those with the standard pump end.

- The new dimension $D_{\max}^*$ for the selected size can be found in the following tables.
- The new dimension L_A^* must be calculated: $L_A^* = L_A - L_G + L_G^*$
 - Dimensions L_A and L_G for the selected size can be found in the standard "Dimensions" table.
 - Dimension L_G^* for the selected size can be found in the following tables.

Table 23: Dimensions, alternative pump ends, UPA S 350 - 500

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve	G8 ¹⁸⁾	336	395
	NPT 8-8	336	395
	DN 200 / PN10, PN16	352	355
	DN 200 / PN25	362	355
	DN 200 / PN40	375	355
	DN 200 / PN63	430	355
	G10 / NPT 10-8	416	680
	DN 250 / PN10	416	630
	DN 250 / PN16	416	630
	DN 250 / PN25	425	630
	DN 250 / PN40	450	630
	DN 250 / PN63	470	630
Connection nozzles	G8 ¹⁸⁾	336	320
	NPT 8-8	336	320
	DN 200 / PN10, PN16	352	265
	DN 200 / PN25	362	265
	DN 200 / PN40	375	265
	DN 200 / PN63	430	265
	G10 / NPT 10-8	416	420
	DN 250 / PN10	416	420
	DN 250 / PN16	416	420
	DN 250 / PN25	425	420
	DN 250 / PN40	450	420
	DN 250 / PN63	470	420

Table 24: Dimensions, alternative pump ends, UPA S 350 - 650

Pump end		$D_{\max}^*$	L_G^*
		[mm]	[mm]
Lift check valve	G8 ¹⁸⁾	358	395
	NPT 8-8	358	395
	DN 200 / PN10, PN16	361	355
	DN 200 / PN25	371	355
	DN 200 / PN40	378	355
	DN 200 / PN63	430	355
	G10 / NPT 10-8	416	640
	DN 250 / PN10	416	590
	DN 250 / PN16	416	590
	DN 250 / PN25	425	590
	DN 250 / PN40	450	590
	DN 250 / PN63	470	590
Connection nozzles	G8 ¹⁸⁾	358	320
	NPT 8-8	358	320
	DN 200 / PN10, PN16	361	265
	DN 200 / PN25	371	265
	DN 200 / PN40	378	265

18 Standard pump end

Pump end		D _{max} *	L _G *
		[mm]	[mm]
Connection nozzles	DN 200 / PN63	430	265
	G10 / NPT 10-8	416	380
	DN 250 / PN10	416	380
	DN 250 / PN16	416	380
	DN 250 / PN25	425	380
	DN 250 / PN40	450	380
	DN 250 / PN63	470	380

Scope of supply

Depending on the model, the following items are included in the scope of supply:

- Pump set with motor lead
- Back-up name plate

Optional:

- Extension cable connected or supplied but not fitted
- Cable connector
- Cable clips
- Pedestals

- Cooling shroud, suction shroud or pressure shroud
- Supporting clamps and mounting clamps
- Electrical protection equipment
- Automatic control units
- Centring device
- Spare parts
- Hydraulic performance test
- Frequency inverter
- Voltage filter
- Dry running protection device

Cross sectional drawings with list of components

UPA S 350-500/650, standard design

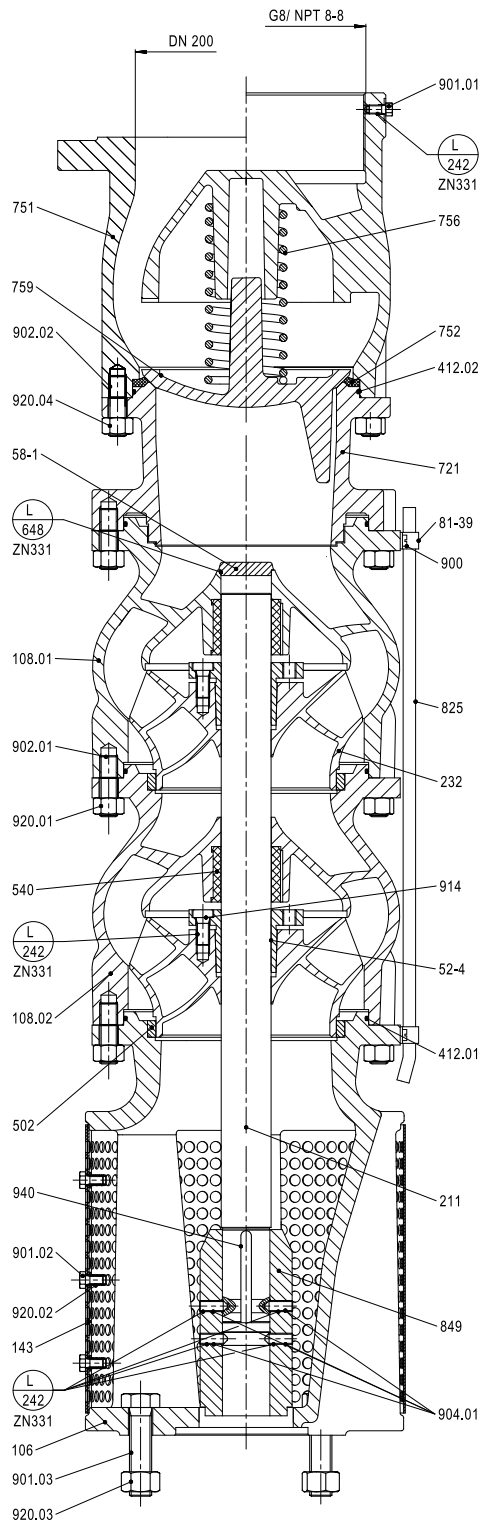


Fig. 8: UPA S 350-500/650 incl. lift check valve G8/NPT 8-8/DN 200, for connection to UMA 250 or UMA-S 250 motor, standard design

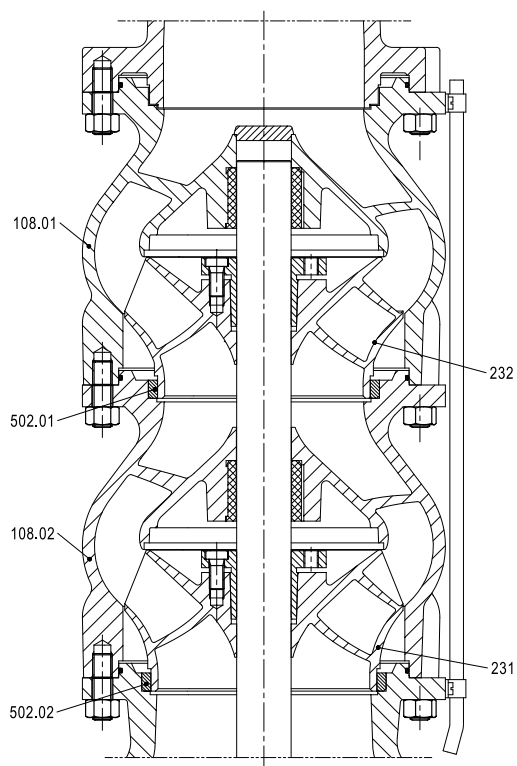


Fig. 9: UPA S 350-650, standard design

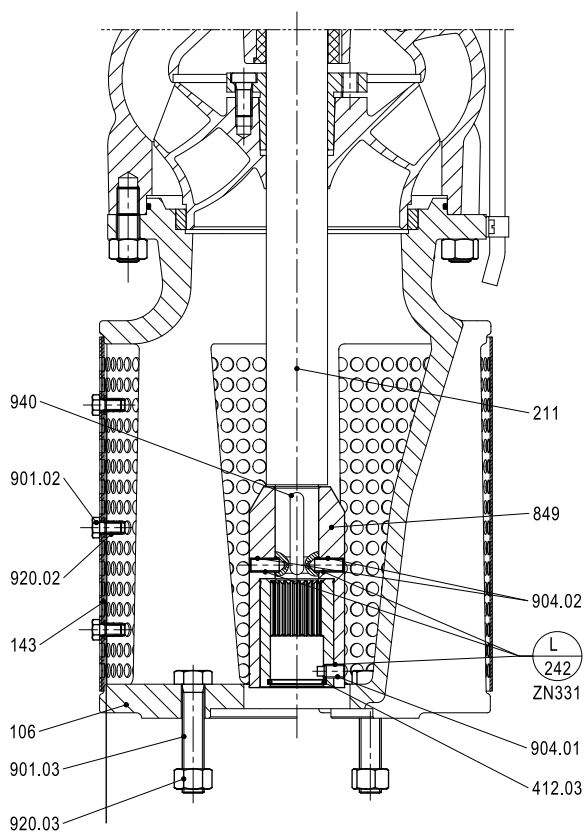


Fig. 10: UPA S 350-500/650 with connection for UMA 200 or UMA-S 200 motor

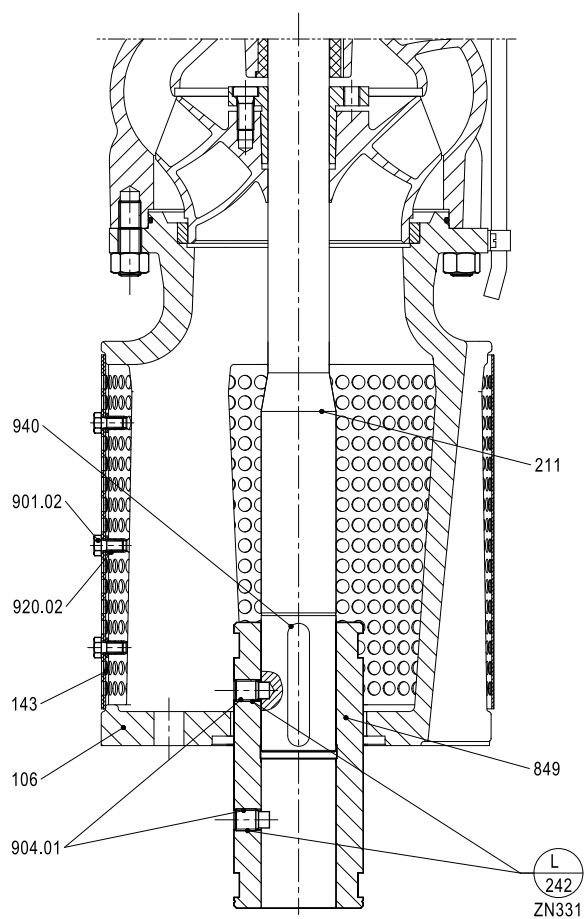


Fig. 11: UPA S 350-500/650 with connection for UMA 300 motor

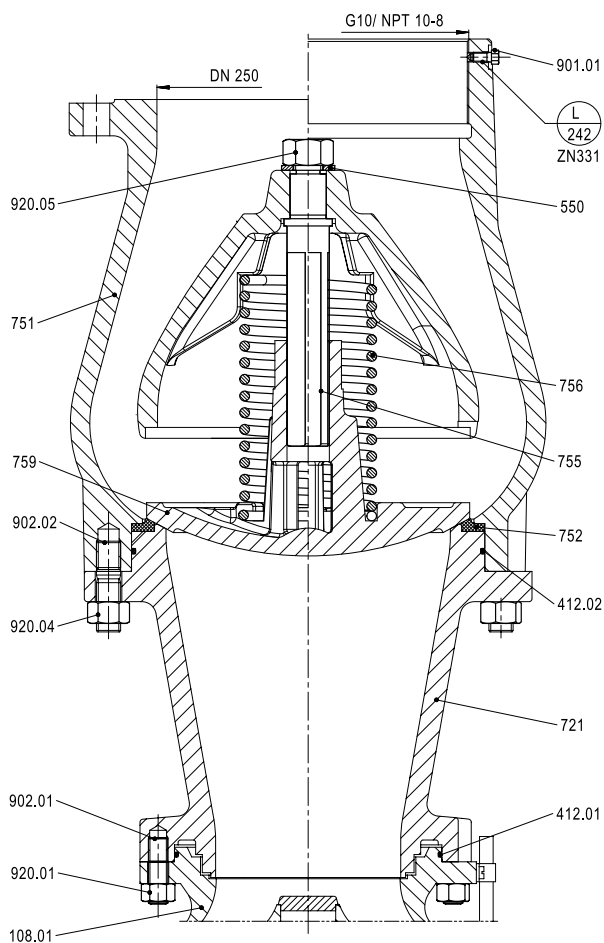


Fig. 12: UPA S 300-500/650, lift check valve G10/NPT10-8/DN 250

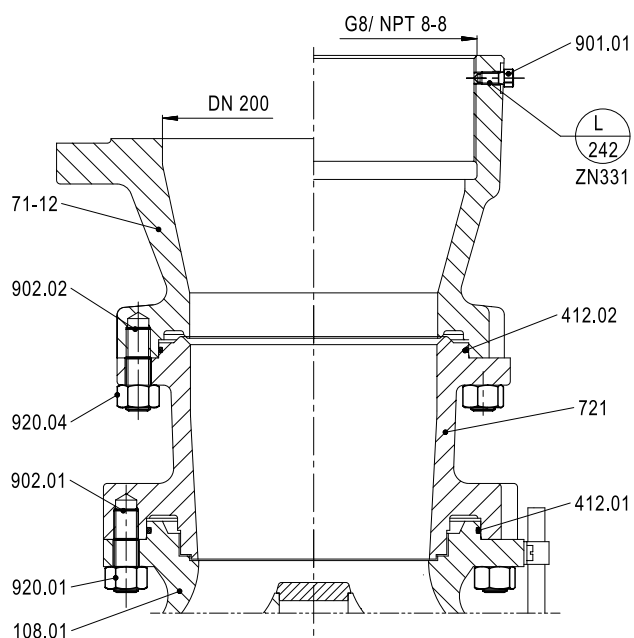


Fig. 13: UPA S 350-500/650, connection nozzle G8/NPT 8-8/DN 200

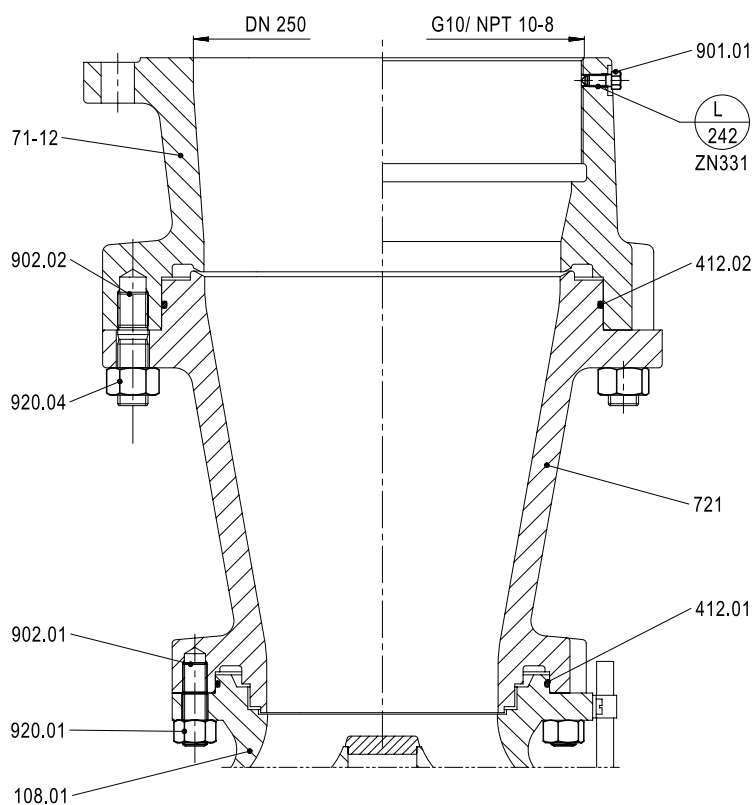


Fig. 14: UPA S 350-500/650, connection nozzle G10/NPT 10-8/DN 250

Table 25: Symbols key

Symbol	Description
	Always secure screw/bolt/stud connections marked with this symbol with Loctite 242 .
	Always secure connections marked with this symbol with Loctite 648 .

Table 26: List of components for UPA S 350, standard design

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1	106	Suction casing	✓	✓	-	106 + 143, 412.01, 502 ¹⁹⁾ / 502.02 ²⁰⁾ , 901.02, 901.03, 920.02, 920.03
			-	-	✓	106 + 143, 412.01, 502 ¹⁹⁾ / 502.02 ²⁰⁾ , 901.02, 920.02
1	108.01	Stage casing (last stage)	✓	✓	✓	108.01 + 412.01, 540, 58-1, 902.01, 920.01
1 per stage -1	108.02	Stage casing	✓	✓	✓	108.02 + 412.01, 502 ¹⁹⁾ / 502.01 ²⁰⁾ , 540, 902.01, 920.01
1	143 ²¹⁾	Suction strainer	✓	✓	✓	143 + 901.02, 920.02
1	211	Pump shaft	✓	-	-	211 + 412.03, 540, 849, 904.01, 904.02, 940
			-	✓	✓	211 + 540, 849, 904.01, 940
1	231 ²⁰⁾	Suction stage impeller	✓	✓	✓	231 + 52-4, 914
1 per stage -1	232 ²⁰⁾	Clockwise impeller	✓	✓	✓	232 + 52-4, 914
1 per stage	232 ¹⁹⁾	Clockwise impeller	✓	✓	✓	232 + 52-4, 914
1 per stage +1	412.01 ²¹⁾	O-ring	✓	✓	✓	-
1	412.02 ²¹⁾	O-ring	✓	✓	✓	-
1	412.03 ²¹⁾	O-ring	✓	-	-	-
1 per stage	502 ²¹⁾¹⁹⁾	Casing wear ring	✓	✓	✓	-
1 per stage -1	502.01 ²¹⁾²⁰⁾	Casing wear ring	✓	✓	✓	-
1	502.02 ²¹⁾²⁰⁾	Casing wear ring	✓	✓	✓	-
1 per stage	52-4 ²¹⁾	Locking sleeve	✓	✓	✓	52-4 + 914
1 per stage	540 ²¹⁾	Bush (EPDM)	✓	✓	✓	-
1	550	Disc	✓	✓	✓	-
1	58-1	Protecting plug	✓	✓	✓	-
1	71-12	Connection nozzle, flanged	✓	✓	✓	71-12 + 902.02, 920.04
		Connection nozzle, threaded	✓	✓	✓	71-12 + 901.01, 902.02, 920.04
1	721	Adapter	✓	✓	✓	721 + 412.02, 902.01, 920.01
1	751 ²²⁾	Valve body, flanged	✓	✓	✓	751 + 752, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	751 + 752, 756, 759, 901.01, 902.02, 920.04
	751 ²³⁾	Valve body, flanged	✓	✓	✓	751 + 550, 752, 755, 756, 759, 902.02, 920.04, 920.05
		Valve body, threaded	✓	✓	✓	751 + 550, 752, 755, 756, 759, 901.01, 902.02, 920.04, 920.05
1	752 ²¹⁾	Valve seat (EPDM)	✓	✓	✓	-
1	755	Valve stem	✓	✓	✓	-
1	759	Valve disc	✓	✓	✓	759 + 550 ²³⁾ , 752, 755 ²³⁾ , 756, 920.05 ²³⁾
2 per electric cable	81-39	Clamp	✓	✓	✓	-
1 per electric cable	825	Cable guard	✓	✓	✓	-
1	849	Sleeve coupling	✓	-	-	849 + 412.03, 904.01, 904.02, 940
			-	✓	✓	849 + 904.01, 940
4 per electric cable	900.01	Hexagon socket head cap screw	✓	✓	✓	-
2	901.01	Hexagon head bolt	✓	✓	✓	-

19 UPA S 350-500 only

20 UPA S 350-650 only

21 Recommended spare part

22 Only for G8, NPT8-8 and DN 200 valve bodies

23 Only for G10, NPT10-8 and DN 250 valve bodies

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
3	901.02	Hexagon head bolt	✓	✓	✓	-
4	901.03	Hexagon head bolt	✓	✓	-	-
(1 per stage +1) × 12	902.01	Stud	✓	✓	✓	-
12	902.02	Stud	✓	✓	✓	-
1	904.01	Grub screw	✓	-	-	-
4		Grub screw	-	✓	-	-
2		Grub screw	-	-	✓	-
2	904.02	Grub screw	✓	-	-	-
1 per stage × 4	914	Hexagon socket head cap screw	✓	✓	✓	-
(1 per stage +1) × 12	920.01	Nut	✓	✓	✓	-
3	920.02	Nut	-	✓	✓	-
4	920.03	Nut	✓	✓	-	-
12	920.04	Nut	✓	✓	✓	-
1	920.05	Nut	✓	✓	✓	-
1	940 ²¹⁾	Key	✓	✓	✓	-

UPA S 350-500/650, wear-resistant version

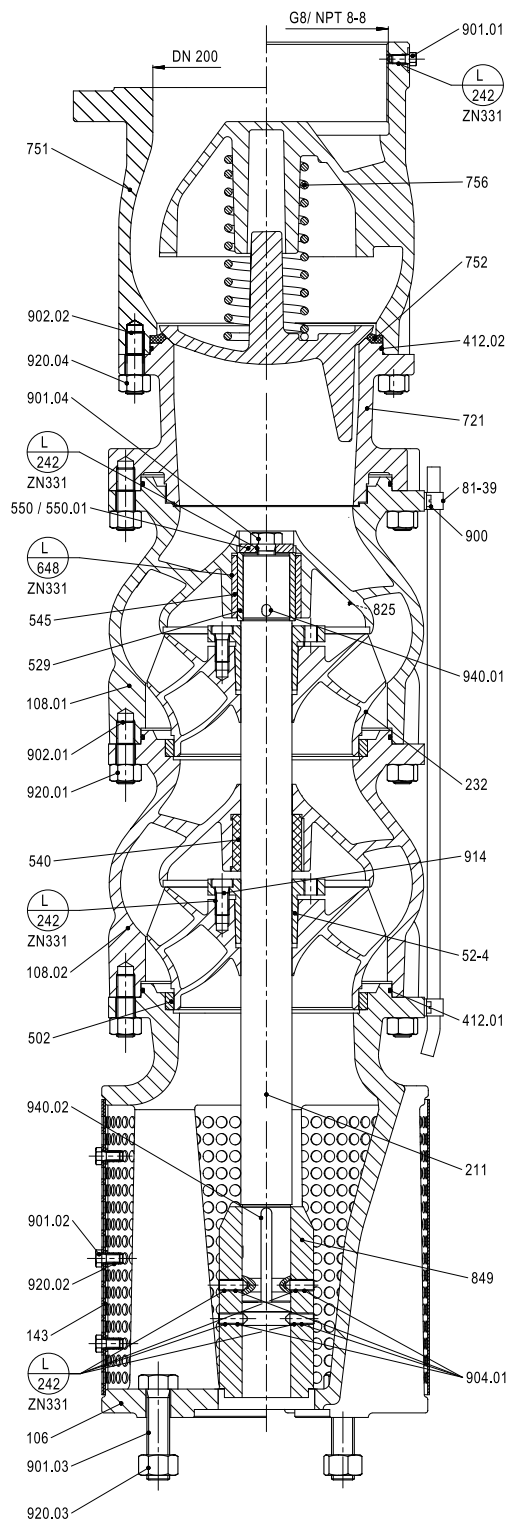


Fig. 15: UPA S 350-500/650 incl. lift check valve G8/NPT 8-8/DN 200, for connection to UMA 250 or UMA-S 250 motor, wear-resistant version

i The following illustrations are identical with those of the standard model:

- UPA S 350-500/650 with connection for UMA 200, UMA-S 200 and UMA 300 motor
- UPA S 350-500/650 with lift check valve and connection nozzle G10/NPT 10-8/DN 250
- UPA S 350-500/650 with lift check valve and connection nozzle G8/NPT 8-8/DN 200

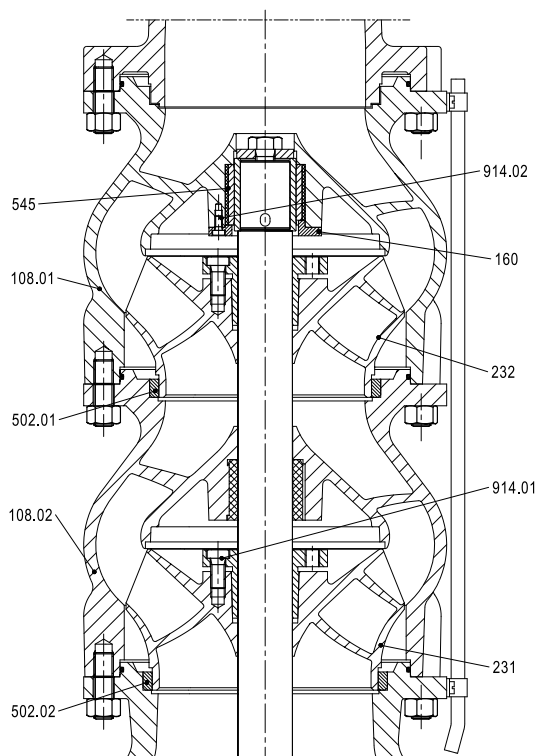


Fig. 16: UPA S 350-650, wear-resistant version

Table 27: Symbols key

Symbol	Description
	Always secure screw/bolt/stud connections marked with this symbol with Loctite 242 .
	Always secure connections marked with this symbol with Loctite 648 .

Table 28: List of components for UPA S 350, wear-resistant version

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1	106	Suction casing	✓	✓	-	106 + 143, 412.01, 502 ²⁴⁾ / 502.02 ²⁵⁾ , 901.02, 901.03, 920.02, 920.03
			-	-	✓	106 + 143, 412.01, 502 ²⁴⁾ / 502.02 ²⁵⁾ , 901.02, 920.02
1	108.01	Stage casing (last stage)	✓	✓	✓	108.01 + 160 ²⁵⁾ , 412.01, 529, 545, 902.01, 914.02 ²⁵⁾ , 920.01
1 per stage -1	108.02	Stage casing	✓	✓	✓	108.02 + 412.01, 502 ²⁴⁾ / 502.01 ²⁵⁾ , 540, 902.01, 920.01
1	143 ²⁶⁾	Suction strainer	✓	✓	✓	143 + 901.02, 920.02
1	160	Cover	✓	✓	✓	160 + 914.02
1	211	Pump shaft	✓	-	-	211 + 412.03, 529, 540, 545, 550.01, 849, 901.04, 904.01, 904.02, 940.01, 940.02

24 UPA S 350-500 only

25 UPA S 350-650 only

26 Recommended spare part

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1	211	Pump shaft	-	✓	✓	211 + 529, 540, 545, 550.01, 849, 901.04, 904.01, 940.01, 940.02
1	231 ⁽²⁵⁾	Suction stage impeller	✓	✓	✓	231 + 52-4, 914.01
1 per stage -1	232 ⁽²⁵⁾	Clockwise impeller	✓	✓	✓	232 + 52-4, 914.01
1 per stage	232 ⁽²⁴⁾	Clockwise impeller	✓	✓	✓	232 + 52-4, 914
1 per stage +1	412.01 ⁽²⁶⁾	O-ring	✓	✓	✓	-
1	412.02 ⁽²⁶⁾	O-ring	✓	✓	✓	-
1	412.03 ⁽²⁶⁾	O-ring	✓	-	-	-
1 per stage	502 ⁽²⁶⁾⁽²⁴⁾	Casing wear ring	✓	✓	✓	-
1 per stage -1	502.01 ⁽²⁶⁾⁽²⁵⁾	Casing wear ring	✓	✓	✓	-
1	502.02 ⁽²⁶⁾⁽²⁵⁾	Casing wear ring	✓	✓	✓	-
1 per stage	52-4 ⁽²⁶⁾	Locking sleeve	✓	✓	✓	52-4 + 914 ⁽²⁴⁾ / 914.01 ⁽²⁵⁾
1	529	Bearing sleeve	✓	✓	✓	529 + 545
1 per stage -1	540 ⁽²⁶⁾	Bush (EPDM)	✓	✓	✓	-
1	545 ⁽²⁶⁾	Bearing bush	✓	✓	✓	545 + 529
1	550.01	Disc	✓	✓	✓	-
1	550.02	Disc	✓	✓	✓	-
1	71-12	Connection nozzle, flanged	✓	✓	✓	71-12 + 902.02, 920.04
		Connection nozzle, threaded	✓	✓	✓	71-12 + 901.01, 902.02, 920.04
1	721	Adapter	✓	✓	✓	721 + 412.02, 902.01, 920.01
1	751 ⁽²⁷⁾	Valve body, flanged	✓	✓	✓	751 + 752, 756, 759, 902.02, 920.04
		Valve body, threaded	✓	✓	✓	751 + 752, 756, 759, 901.01, 902.02, 920.04
	751 ⁽²⁸⁾	Valve body, flanged	✓	✓	✓	751 + 550.02, 752, 755, 756, 759, 902.02, 920.04, 920.05
		Valve body, threaded	✓	✓	✓	751 + 550.02, 752, 755, 756, 759, 901.01, 902.02, 920.04, 920.05
1	752 ⁽²⁶⁾	Valve seat (EPDM)	✓	✓	✓	-
1	755	Valve stem	✓	✓	✓	-
1	756 ⁽²⁶⁾	Valve spring	✓	✓	✓	-
1	759	Valve disc	✓	✓	✓	759 + 550.02 ⁽²⁸⁾ , 752, 755 ⁽²⁸⁾ , 756, 920.05 ⁽²⁸⁾
2 per electric cable	81-39	Clamp	✓	✓	✓	-
1 per electric cable	825	Cable guard	✓	✓	✓	-
1	849	Sleeve coupling	✓	-	-	849 + 412.03, 904.01, 904.02, 940.02
			-	✓	✓	849 + 904.01, 940.02
4 per electric cable	900	Hexagon socket head cap screw	✓	✓	✓	-
2	901.01	Hexagon head bolt	✓	✓	✓	-
3	901.02	Hexagon head bolt	✓	✓	✓	-
4	901.03	Hexagon head bolt	✓	✓	-	-
1	901.04	Hexagon head bolt	✓	✓	✓	-
(1 per stage +1) × 12	902.01	Stud	✓	✓	✓	-
12	902.02	Stud	✓	✓	✓	-
1	904.01	Grub screw	✓	-	-	-
4		Grub screw	-	✓	-	-
2		Grub screw	-	-	✓	-
2	904.02	Grub screw	✓	-	-	-

- 27** Only for G8, NPT8-8 and DN 200 valve bodies
28 Only for G10, NPT10-8 and DN 250 valve bodies

Quantity	Part No.	Description	Motor size			Scope of supply of spare parts kit
			8 inch UMA 200, UMA-S 200	10 inch UMA 250, UMA-S 250	12 inch UMA 300	
1 per stage × 4	914 ²⁴⁾ / 914.01 ²⁵⁾	Hexagon socket head cap screw	✓	✓	✓	-
4	914.02	Hexagon socket head cap screw	✓	✓	✓	-
(1 per stage +1) × 12	920.01	Hexagon nut	✓	✓	✓	-
3	920.02	Hexagon nut	-	✓	✓	-
4	920.03	Hexagon nut	✓	✓	-	-
12	920.04	Hexagon nut	✓	✓	✓	-
1	920.05	Hexagon nut	✓	✓	✓	-
1	940.01 ²⁶⁾	Key	✓	✓	✓	-
1	940.02 ²⁶⁾	Key	✓	✓	✓	-

Accessories

Installation parts

Table 29: Selection table

Description		Mat. No.	Weight [kg]
Set of pedestals for horizontal installation; comprising: 2 pedestals and fastening elements			
1.4401	UPA S 350-500 with UMA 200 or UMA-S 200 motor	05317622	9
	UPA S 350-500 with UMA 250 or UMA-S 250 motor	05317624	17.2
	UPA S 350-500 with UMA 300-250/2 motor with support frame	05317626	79.1
	UPA S 350-500 with UMA 300-300/2 motor with support frame	05317628	85.1
	UPA S 350-650 with UMA 200 or UMA-S 200 motor	05100031	9
	UPA S 350-650 with UMA 250 or UMA-S 250 motor	05100034	17.2
	UPA S 350-650 with UMA 300-250/2 motor with support frame	05100046	79.1
	UPA S 350-650 with UMA 300-300/2 motor with support frame	05100047	85.1
1.4539	UPA S 350-500 with UMA 200 or UMA-S 200 motor	05317623	9
	UPA S 350-500 with UMA 250 or UMA-S 250 motor	05317625	17.2
	UPA S 350-500 with UMA 300-250/2 motor with support frame	05317627	79.1
	UPA S 350-500 with UMA 300-300/2 motor with support frame	05317629	85.1
	UPA S 350-650 with UMA 200 or UMA-S 200 motor	05073102	9
	UPA S 350-650 with UMA 250 or UMA-S 250 motor	05073105	17.2
	UPA S 350-650 with UMA 300-250/2 motor with support frame	05073135	79.1
	UPA S 350-650 with UMA 300-300/2 motor with support frame	05073136	85.1
Bolts for fastening the pedestals to the foundation are not included in the scope of supply; they must be provided at the site.			
Suction shroud and pressure shroud		On request	-
Adapter		On request	-
Pair of supporting clamps and installation clamps for the following riser sizes, incl. fastening elements, material: painted steel			
R 6 / DN 150	L = 800 mm, F = 31,0 kN	95000304	29
R 8 / DN 200	L = 900 mm, F = 108 kN	95000307	70
R 10 / DN 250	L = 1000 mm, F = 108 kN	95000309	78
2 pairs are required for installation and dismantling.			

Centring device

Table 30: Centring device overview

	Motor size	Well diameter	Mat. No.			Weight
			Material variant			
		[mm]	Plastic	Galvanised steel	Stainless steel	[kg]
	UMA 150	250-299	90047662	-	-	-
	UMA-S 150	300-349	90047663	-	-	-
		350-400	90047664	-	-	-
		UMA 200	300-349	90047665	-	-
	UMA-S 200	350-400	90047666	-	-	-
	UMA-S 250	450-499	-	90037033	90037039	-
	UMA-S 250	500-549	-	90034400	90037041	-
		550-599	-	90037035	90037043	-
		600-650	-	90037037	90037045	-

Cooling shroud

Table 31: Cooling shroud overview

UPA S 350-...	Installation		Motor		Cooling shroud					Strainer				
	Vertical	Horizontal			Ø × length	Weight	Material			Ø × length	Weight	Material		
			1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L	1.4301/ AISI304SS	1.4401/ AISI316SS			1.4539/ AISI904L		
			Type	P _N										

Flow velocity past the motor

Application

- Required motor cooling

Flow velocity past the motor

$v = 0 \text{ m/s}$

- The flow velocity past the motor is undefined.
- A free flow of heat is established. This heat flow must not be influenced or hindered by any structures, and a supply of fresh water must be provided at all times.
 - Example: pump set freely suspended in a vertical position inside a large tank

Flow velocity past the motor

$v > 0.2 \text{ m/s}$

$v > 0.5 \text{ m/s}$

- The flow velocity past the motor is defined for the installation conditions, see table below.
Decisive factors are the inside well diameter or the inside shroud diameter, the operating data and the outside dimensions of the pump set.
 - Example: pump set installed in a vertical position above the screen/filter in a well; values in accordance with the table below.
 - Example: pump set installed in a horizontal position in a tank, fitted with a cooling shroud, or pump set installed in a vertical position in a pump sump, fitted with a cooling shroud; values in accordance with the table below.

Table 32: Maximum permissible well diameter or inside shroud diameter

Flow rate	Flow velocity	Well diameter or inside shroud diameter [mm]				
Q [m³/h]	v [m/s]	UMA 150 UMA-S 150	UMA 200 UMA-S 200	UMA 250 UMA-S 250	UMA 300	14D
15	$\geq 0,2$	≤ 215	-	-	-	-
	$\geq 0,5$	≤ 175	-	-	-	-
25	$\geq 0,2$	≤ 255	-	-	-	-
	$\geq 0,5$	≤ 195	-	-	-	-
50	$\geq 0,2$	≤ 330	≤ 350	-	-	-
	$\geq 0,5$	≤ 235	≤ 265	-	-	-
75	$\geq 0,2$	≤ 390	≤ 410	≤ 430	-	-
	$\geq 0,5$	≤ 270	≤ 300	≤ 330	-	-
100	$\geq 0,2$	≤ 445	≤ 460	≤ 480	-	-
	$\geq 0,5$	≤ 300	≤ 325	≤ 355	-	-
125	$\geq 0,2$	≤ 490	≤ 510	≤ 525	-	-
	$\geq 0,5$	≤ 330	≤ 350	≤ 380	-	-
150	$\geq 0,2$	≤ 535	≤ 550	≤ 565	≤ 590	-
	$\geq 0,5$	≤ 355	≤ 380	≤ 400	≤ 430	-
175	$\geq 0,2$	≤ 575	≤ 590	≤ 605	≤ 625	-
	$\geq 0,5$	≤ 380	≤ 400	≤ 420	≤ 450	-
200	$\geq 0,2$	≤ 615	≤ 625	≤ 640	≤ 660	≤ 690
	$\geq 0,5$	≤ 405	≤ 420	≤ 445	≤ 470	≤ 510
250	$\geq 0,2$	≤ 680	≤ 690	≤ 705	≤ 725	≤ 750
	$\geq 0,5$	≤ 445	≤ 460	≤ 480	≤ 505	≤ 540
300	$\geq 0,2$	≤ 745	≤ 755	≤ 765	≤ 780	≤ 800
	$\geq 0,5$	≤ 485	≤ 500	≤ 515	≤ 540	≤ 570
350	$\geq 0,2$	-	≤ 810	≤ 820	≤ 835	≤ 860
	$\geq 0,5$	-	≤ 530	≤ 550	≤ 570	≤ 600
400	$\geq 0,2$	-	≤ 865	≤ 875	≤ 890	≤ 910
	$\geq 0,5$	-	≤ 565	≤ 580	≤ 605	≤ 630
500	$\geq 0,2$	-	≤ 960	≤ 970	≤ 985	≤ 1000
	$\geq 0,5$	-	≤ 625	≤ 640	≤ 660	≤ 690
600	$\geq 0,2$	-	≤ 1050	≤ 1055	≤ 1070	≤ 1090
	$\geq 0,5$	-	≤ 680	≤ 695	≤ 710	≤ 740
800	$\geq 0,2$	-	≤ 1205	≤ 1215	≤ 1225	≤ 1240
	$\geq 0,5$	-	≤ 775	≤ 790	≤ 805	≤ 830
1000	$\geq 0,2$	-	≤ 1345	≤ 1350	≤ 1360	≤ 1370
	$\geq 0,5$	-	≤ 865	≤ 875	≤ 890	≤ 910

Flow rate	Flow velocity	Well diameter or inside shroud diameter [mm]				
Q [m³/h]	v [m/s]	UMA 150 UMA-S 150	UMA 200 UMA-S 200	UMA 250 UMA-S 250	UMA 300	14D
1200	≥ 0,2	-	-	-	≤ 1485	≤ 1500
	≥ 0,5	-	-	-	≤ 965	≤ 980
1400	≥ 0,2	-	-	-	≤ 1600	≤ 1610
	≥ 0,5	-	-	-	≤ 1030	≤ 1050
1600	≥ 0,2	-	-	-	≤ 1705	≤ 1720
	≥ 0,5	-	-	-	≤ 1100	≤ 1120
1800	≥ 0,2	-	-	-	≤ 1805	≤ 1820
	≥ 0,5	-	-	-	≤ 1165	≤ 1180
2000	≥ 0,2	-	-	-	≤ 1900	≤ 1910
	≥ 0,5	-	-	-	≤ 1225	≤ 1240

Power cables

Main applications

- For use with drinking water applications
- Motor lead for submersible motors, laid under water
- Extension cable for the motor lead, laid freely exposed to air and in contact with surfaces
- Unshielded and shielded cable versions

Operating data

Table 33: Operating properties

Characteristic		Value
Rated voltage	V_N [V]	≤ 1000
Ambient temperature	T [°C]	≤ 50
Immersion depth	IM [m]	≤ 500

Designation

Unshielded cable

Example: ZN 1391 - G FL GWT - J 4G25 - Cu-rubber

Table 34: Designation key for unshielded cable

Abbreviation	Description
ZN	Company standard
G	Insulation, rubber
FL	Type of power cable
	FL Flat
	RD Round
GWT	Suitable for drinking water
J	Information about earth conductor
	J With earth conductor
	O Without earth conductor
4G25	Number of cores
	4G25 With earth conductor, 4-core, cross-section 25 mm ²
	3 × 25 Without earth conductor, 3-core, cross-section 25 mm ²
Cu-rubber	Material

Shielded cable

Example: ZN 1391 - S 07 B C4 B - F 3X6 KON

Table 35: Designation key for shielded cable

Abbreviation	Description
ZN	Company standard
S	Basic type according to manufacturer
07	Rated voltage U_0/U
	05 300/500 V
	07 450/750 V
	1/10 600/1000 V
B	Insulation material
	B EPR
C4	Braided shield
B	Sheath material
	B EPR
F	Flexible

Abbreviation	Description
3X6	Nominal cross-section per phase
	3 × 6 Without earth conductor, 3-core, cross-section 6 mm ²
	4G35 With earth conductor, 4-core, cross-section 35 mm ²
KON	Earth conductor design
	KON Earth conductor, concentric as braided shield
	3G16/3 Earth conductor cross-section = nominal phase cross-section (earth conductor divided into three cores, earth conductor cross-section = 16 mm ²), G = with earth conductor

Design details

Unshielded rubber-sheathed cable

Comprising:

- Conductor
 - Copper
 - Bare
 - Fine-stranded
 - Class 5 to DIN EN 60228
- Insulation
 - EPR-based insulation compound (ethylene-propylene rubber) resistant to ozone, weather and water
- Core identification
 - Core colours to DIN EN 50525-1
- Sheath
 - Core insulation made of EPR
 - Outer sheath made of EPDM rubber compound
 - Colour: blue

Shielded rubber-sheathed cable

Comprising:

- Conductor
 - Copper
 - Bare
 - Fine-stranded
 - Class 5 to DIN VDE 0295 / IEC 60228
- Insulation
 - EPR-based insulation compound (ethylene-propylene rubber) resistant to ozone, weather and water
- Core identification
 - Core colours to DIN VDE 0293-308:2003
- Two-layer sheath design
 - Inner sheath: Special EPR compound as per German KTW recommendations, colour: blue
 - Outer sheath: Special EPR compound as per German KTW recommendations, DVGW (Worksheet 270), WRAS and ACS, colour: blue
- Shielding
 - Tinned copper wire braiding over the inner layer of the sheath system, max. coupling resistance 30 Ohm/km at 30 MHz
- Marking
 - HYDROFIRM(T) EMV-FC S1BC4B-F 3X10/10 KON 0,6/1 kV KTW DVGW W270 ACS WRAS

Table 36: Overview of electric cables

Type of electric cable	Characteristic
Flat cable, unshielded	
	3 cores, flat
	4 cores, flat
Round cable, unshielded	
	1 core, round
	4 cores, round
Round cable, shielded	
	4 cores, round

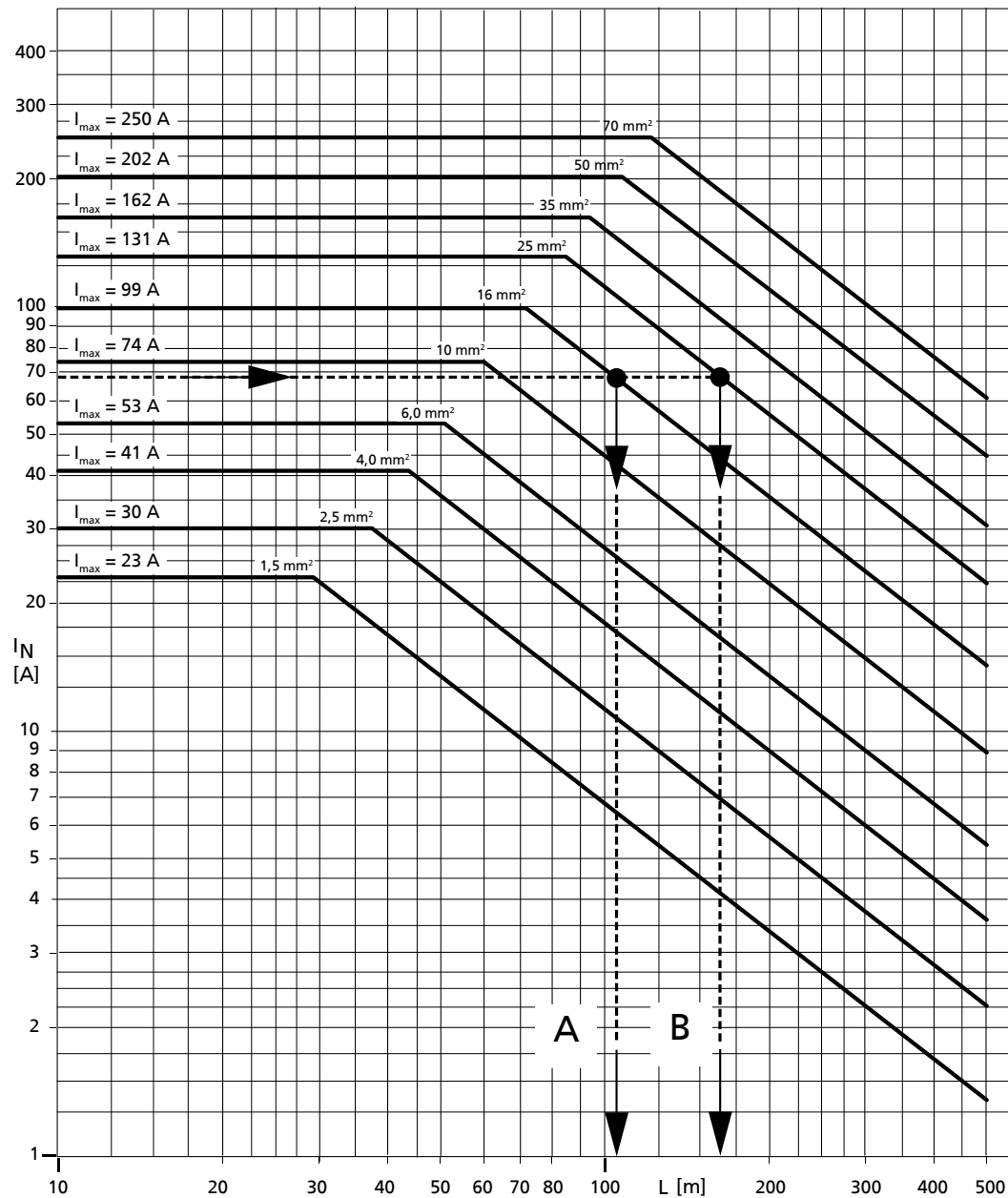
Selection information

Permissible cable lengths for starting method: DOL

Applies to 1 cable or 2 cables in parallel²⁹⁾

Conditions:

- $V = 400 \text{ V}$; $\Delta V = 3 \%$; $T \leq 30 \text{ °C}$; laid freely exposed to air and in contact with surfaces



Example A

$I_N = 68 \text{ A}$
Cable design: 1 x 16 mm^2
Cable length: $L \leq 105 \text{ m}$

Example B

$I_N = 68 \text{ A}$
Cable design: 1 x 25 mm^2
Cable length: $L \leq 165 \text{ m}$

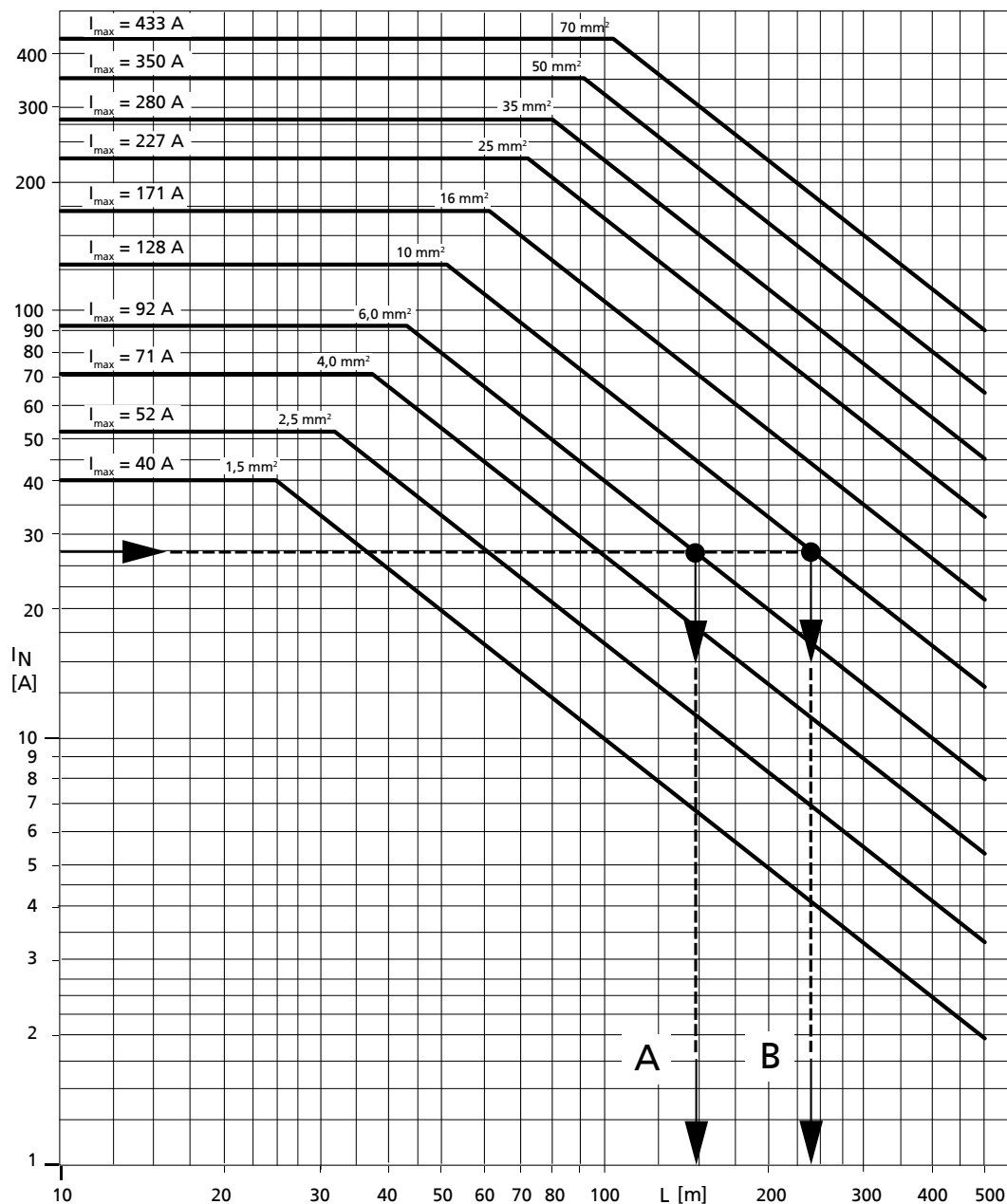
²⁹ For parallel cables the value is doubled to obtain the permissible length.

Permissible cable lengths for starting method: YΔ

Applies to 2 cables

Conditions:

- $V = 400 \text{ V}$; $\Delta V = 3 \%$; $T \leq 30 \text{ }^{\circ}\text{C}$; laid freely exposed to air and in contact with surfaces



Example A

$I_N = 27 \text{ A}$
Cable design: $2 \times 6.0 \text{ mm}^2$
Cable length: $L \leq 145 \text{ m}$

Example B

$I_N = 27 \text{ A}$
Cable design: $2 \times 10 \text{ mm}^2$
Cable length: $L \leq 235 \text{ m}$

Voltage drop in the extension cable

When calculating the required conductor cross-section q not only I_N , but also the voltage drop ΔV along the cable length L (distance from motor to control unit) must be taken into account. Proper functioning of our submersible motors requires $\Delta V \leq 3 \%$ of the

supply voltage V .

If $\Delta V > 3 \%$, a larger conductor cross-section must be used. The voltage drop is calculated using the following equations:

Starting method DOL/autotransformer

- 1 cable:

$$\Delta U = \frac{3,1 \times L \times I_N \times \cos \varphi}{q \times U} \quad [\%]$$

- 2 cables in parallel (II):

$$\Delta U = \frac{1,55 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

Starting method YΔ (2 cables):

$$\Delta U = \frac{2,1 \times L \times I_N \times \cos \varphi}{q \times U} [\%]$$

Power loss ΔP:

$$\Delta P = \frac{\Delta U}{(\cos \varphi)^2} [\%]$$

Table 37: Key

Code	Description
L	Single cable length [m]
I _N	Rated current [A]:
cos φ	Power factor at 4/4 load
q	Conductor cross-section [mm ²]
U	Supply voltage [V]

Maximum permissible rated motor current
Table 38: For ambient temperature t ≤ 30 °C

Starting method	Used as ...	I _{max} [A] for the following conductor cross-sections [mm ²]										
		1,5	2,5	4	6	10	16	25	35	50	70	95
DOL (1 cable or 2 cables in parallel)	Motor lead	29	38	52	67	94	125	166	205	256	316	517
	Extension cable	23	30	41	53	74	99	131	162	202	250	409
YΔ (2 cables)	Motor lead	50	66	90	116	163	217	288	355	443	547	895
	Extension cable	40	52	71	92	128	171	227	280	350	433	708

Dimensions and weights
Table 39: Technical data, unshielded electric extension cable for connection with the motor lead

Core type	Description	Cable cross-section	Height	Width	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]	[mm]	[mm]	[mm]				[kg/m]
●●●	Unshielded electric extension cable, 3-core, flat, without earth conductor	1,5	5,2 ^{+1,0}	11,0 ^{+2,0}	-	G FL GWT-O	3×1,5	90068148	0.11
		2,5	6,1 ^{+1,5}	13,2 ^{+2,3}	-	G FL GWT-O	3×2,5	90068149	0.17
		4	7,0 ^{+2,0}	15,5 ^{+3,5}	-	G FL GWT-O	3×4	90068150	0.25
		6	7,6 ^{+2,2}	17,4 ^{+3,6}	-	G FL GWT-O	3×6	90068151	0.32
		10	9,3 ^{+2,2}	21,5 ^{+3,5}	-	G FL GWT-O	3×10	90068152	0.49
		16	11,2 ^{+2,3}	26,7 ^{+4,3}	-	G FL GWT-O	3×16	90068153	0.75
		25	13,0 ^{+2,5}	31,6 ^{+3,9}	-	G FL GWT-O	3×25	90068154	1.11
		35	14,6 ^{+2,9}	35,5 ^{+5,0}	-	G FL GWT-O	3×35	90068155	1.44
		50	17,0 ^{+3,0}	42,1 ^{+4,9}	-	G FL GWT-O	3×50	90068156	2.05
		70	19,3 ^{+2,7}	48,4 ^{+3,6}	-	G FL GWT-O	3×70	90068157	2.76
		95	-	-	-	-	-	-	-
●●●●	Unshielded electric extension cable, 4-core, flat, with earth conductor	1,5	5,2 ^{+1,0}	14,5 ^{+2,7}	-	G FL GWT-J	4G×1,5	90068158	0.17
		2,5	6,1 ^{+1,5}	17,5 ^{+2,5}	-	G FL GWT-J	4G×2,5	90068159	0.24
		4	-	-	-	G FL GWT-J	4G×4	90068160	0.34
		6	7,6 ^{+2,2}	23,5 ^{+3,0}	-	G FL GWT-J	4G×6	90068161	0.44
		10	9,3 ^{+2,2}	29,0 ^{+3,5}	-	G FL GWT-J	4G×10	90068162	0.7
		16	11,2 ^{+2,3}	35,0 ^{+2,4}	-	G FL GWT-J	4G×16	90068163	1.03
		25	13,0 ^{+2,5}	41,5 ^{+4,5}	-	G FL GWT-J	4G×25	90068164	1.46
		35	-	-	-	G FL GWT-J	4G×25	90068165	1.96
		50	-	-	-	-	-	-	-
		70	-	-	-	-	-	-	-
		95	-	-	-	-	-	-	-
●	Unshielded electric extension cable, 1-core, round, without earth conductor	1,5	-	-	5,3 ^{+1,1}	G RD GWT-O	1×1,5	01046306	0.01
		2,5	-	-	-	-	-	-	-
		4	-	-	-	-	-	-	-
		6	-	-	7,4 ^{+1,2}	G RD GWT-O	1×6	01042783	0.11

Core type	Description	Cable cross-section	Height	Width	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]							
	Unshielded electric extension cable, 1-core, round, without earth conductor	10	-	-	9,1 ^{+3,4}	G RD GWT-O	1x10	01042784	0.18
		16	-	-	10,7 ^{+2,4}	G RD GWT-O	1x16	01042785	0.27
		25	-	-	12,6 ^{+2,6}	G RD GWT-O	1x25	01042786	0.38
		35	-	-	13,8 ^{+3,6}	G RD GWT-O	1x35	01042787	0.5
		50	-	-	16,2 ^{+3,6}	G RD GWT-O	1x50	01042788	0.7
		70	-	-	18,5 ^{+3,6}	G RD GWT-O	1x70	01042789	0.94
		95	-	-	21,9 ^{+1,5}	G RD GWT-O	1x95	01042794	1.15
	Unshielded electric extension cable, 4-core, round, with earth conductor	1,5	-	-	10,0 ^{+2,0}	G RD GWT-J	4G1,5	90068174	0.18
		2,5	-	-	12,0 ^{+1,9}	G RD GWT-J	4G2,5	90068175	0.26
		4	-	-	13,9 ^{+2,0}	G RD GWT-J	4G4	90068176	0.36
		6	-	-	15,7 ^{+2,1}	G RD GWT-J	4G6	90068177	0.48
		10	-	-	21,1 ^{+2,1}	G RD GWT-J	4G10	90068178	0.84
		16	-	-	24,5 ^{+4,3}	G RD GWT-J	4G16	90068179	1.22
		25	-	-	29,7 ^{+4,3}	G RD GWT-J	4G25	90068180	1.77
		35	-	-	33,3 ^{+5,5}	G RD GWT-J	4G35	90068181	2.3
		50	-	-	39,0 ^{+5,6}	G RD GWT-J	4G50	90068182	3.19
		70	-	-	44,2 ^{+5,8}	G RD GWT-J	4G70	90068183	4.36
		95	-	-	-	-	-	-	-

Table 40: Technical data, shielded electric extension cable for connection with the motor lead

Core type	Description	Cable cross-section	Diameter	Cable designation	Dimensions	Mat. No.	Weight
		[mm ²]					
	Shielded electric extension cable, 4-core, round	0,5	7,5 ^{+1,0}	S 05 B C4 B-F	F4x0,5	05536014	0.09
		6	17,8 ^{+2,0}	S 07 B C4 B-F	3x6/6 KON	05535908	0.5
		16	24,7 ^{+3,0}	S 07 B C4 B-F	3x16+3G16/3	05535909	1.25
		35	32,1 ^{+2,0}	S 07 B C4 B-F	3x35+3G16/3	05536010	2.1
		50	36,8 ^{+2,0}	S 07 B C4 B-F	3x50+3G25/3	05536011	2.78
		70	40,4 ^{+3,0}	S 07 B C4 B-F	3x70+3G35/3	05536012	3.77
		95	45,0 ^{+2,0}	S 07 B C4 B-F	3x95+3G50/3	05536013	4.85

Cable connector

Application

For providing a sealed, water-tight connection between electric extension cables and motor leads of submersible motors.

- Ambient temperature: $T \leq +50\text{ °C}$
- Immersion depth: $\leq 500\text{ m}$

Selection table

The table below only shows the **electric cables on stock**. Cable connectors for other electric cables on request

Table 41: Key

Cable symbols	Explanation	Cable designation
	1-core, round, unshielded	G RD GWT - O 1 G
	4-core, round, unshielded	G RD GWT - J 4 G
	3-core, flat, unshielded	G FL GWT - O 3x ...
	4-core, flat, unshielded	G FL GWT - J 4G ...
	4-core, round, shielded	S 07 B C4 B-F - 4G35

Cable connector for an electric extension cable
Table 42: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm²]										
[mm²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
2,5	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
4	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
6	-	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
	-	-	-	-	-	29F 95005160 ³⁰⁾	29F 95005160 ³⁰⁾	39F 95005510 ³⁰⁾	-	-	-
10	-	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-	-
	-	-	-	-	29F 95005159 ³⁰⁾	29F 95005161 ³⁰⁾	29F 95005161 ³⁰⁾	39F 95005176 ³⁰⁾	-	-	-
16	-	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-
25	-	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-
35	-	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	-
50	-	-	-	-	-	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	-
70	-	-	-	-	-	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	66 95005206 ³⁰⁾ 90053907 ³¹⁾	66 95005207 ³⁰⁾ 90053907 ³¹⁾	-

Table 43: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm²]										
[mm²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
2,5	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
4	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-

3400.515/01-EN

30 MVB - material No. for cable connector, extended and sealed during manufacture

31 OVB - material No. for cable connector, not connected and supplied but not fitted

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
6	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
10	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-	-
16	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90053910 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-
25	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 95005192 ³⁰⁾ 90053913 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-

Table 44: Cable connector: electric extension cable, + or + →

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
+ or +											
1,5	28 90049386 ³⁰⁾ 90049396 ³¹⁾	28 90049386 ³⁰⁾ 90049396 ³¹⁾	28 90049386 ³⁰⁾ 90049396 ³¹⁾	35 95005165 ³⁰⁾ 90039349 ³¹⁾	35 95005165 ³⁰⁾ 90039349 ³¹⁾	-	-	-	-	-	-
2,5	28 90049388 ³⁰⁾ 90049398 ³¹⁾	28 90049388 ³⁰⁾ 90049398 ³¹⁾	28 90049388 ³⁰⁾ 90049398 ³¹⁾	35 95005165 ³⁰⁾ 90039349 ³¹⁾	35 95005165 ³⁰⁾ 90039349 ³¹⁾	-	-	-	-	-	-
4	-	-	35 95005165 ³⁰⁾ 90039349 ³¹⁾	35 95005165 ³⁰⁾ 90039349 ³¹⁾	35 95005165 ³⁰⁾ 90039349 ³¹⁾	43 90049390 ³⁰⁾ 90049400 ³¹⁾	-	-	-	-	-
6	-	-	-	43 90049390 ³⁰⁾ 90049400 ³¹⁾	43 90049390 ³⁰⁾ 90049400 ³¹⁾	43 90049390 ³⁰⁾ 90049400 ³¹⁾	53 90049392 ³⁰⁾ 90049402 ³¹⁾	53 90049392 ³⁰⁾ 90049402 ³¹⁾	66 90033248 ³⁰⁾ 90055737 ³¹⁾	66 90033248 ³⁰⁾ 90055737 ³¹⁾	-
10	-	-	-	66 95005522 ³⁰⁾ 90054294 ³¹⁾	66 95005522 ³⁰⁾ 90054294 ³¹⁾	66 90039412 ³⁰⁾ 90053905 ³¹⁾	66 90039412 ³⁰⁾ 90053905 ³¹⁾	66 90039412 ³⁰⁾ 90053905 ³¹⁾	66 90039412 ³⁰⁾ 90053905 ³¹⁾	66 90033248 ³⁰⁾ 90055737 ³¹⁾	-
16	-	-	-	-	-	66 90039361 ³⁰⁾ 90053906 ³¹⁾	66 90039361 ³⁰⁾ 90053906 ³¹⁾	66 90039361 ³⁰⁾ 90053906 ³¹⁾	66 90036538 ³⁰⁾ 90055738 ³¹⁾	66 90036538 ³⁰⁾ 90055738 ³¹⁾	-
25	-	-	-	-	-	-	78 90039701 ³⁰⁾ 90053908 ³¹⁾	78 90039701 ³⁰⁾ 90053908 ³¹⁾	78 90039701 ³⁰⁾ 90053908 ³¹⁾	78 90039701 ³⁰⁾ 90053908 ³¹⁾	-

Table 45: Cable connector: electric extension cable, →

Motor lead	Electric extension cable [mm ²]										
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70	95
1,5	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	-	-	-	-	-
2,5	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	-	-	-	-	-
4	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	-	-	-	-	-
6	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	28 90049385 ³⁰⁾ 95005106 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	-	-	-
10	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	35 90049387 ³⁰⁾ 90049397 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	-	-
16	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	43 90049389 ³⁰⁾ 90049399 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	66 95005370 ³⁰⁾ 90053914 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	-
25	-	-	-	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	66 95005523 ³⁰⁾ 90053914 ³¹⁾	78 95005527 ³⁰⁾	-
35	-	-	-	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	53 90049391 ³⁰⁾ 90049401 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	78 95005527 ³⁰⁾	-
50	-	-	-	-	-	66 95005370 ³⁰⁾ 90053914 ³¹⁾	66 95005520 ³⁰⁾ 90053914 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	66 95005214 ³⁰⁾ 90053914 ³¹⁾	78 95005527 ³⁰⁾	-
70	-	-	-	-	-	-	-	-	78 95005527 ³⁰⁾	78 95005527 ³⁰⁾	-

Table 46: Cable connector: extension cable for Pt100 sensor, →

Sensor cable for Pt100	Extension cable [mm ²]										
[mm ²]	0,5	-	-	-	-	-	-	-	-	-	-
0,5	18 90062234 ³⁰⁾ 90062233 ³¹⁾	-	-	-	-	-	-	-	-	-	-

Table 47: Cable connector: extension cable for Pt100 sensor, →

Sensor cable for Pt100 [mm²]	Extension cable [mm²]										
	1,5	2,5	4	6	10	16	25	35	50	70	95
35	-	-	-	-	-	-	-	35 95005172 ³⁰⁾ 90023666 ³¹⁾	-	-	-
50	-	-	-	-	-	-	-	-	35 95005173 ³⁰⁾ 90023666 ³¹⁾	-	-
70	-	-	-	-	-	-	-	-	-	35 95005174 ³⁰⁾ 90023666 ³¹⁾	-
95	-	-	-	-	-	-	-	-	-	-	43 95005188 ³⁰⁾

Cable connector for a shielded electric extension cable
Table 48: Cable connector: shielded electric extension cable, →

Motor lead	Shielded electric extension cable [mm²]									
[mm²]	1,5	2,5	4	6	10	16	25	35	50	70
										
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	35 90049387 ³²⁾ 90049397 ³³⁾	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	-	-
4	-	-	-	35 90049387 ³²⁾ 90049397 ³³⁾	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	-	-
6	-	-	-	35 90049387 ³²⁾ 90049397 ³³⁾	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	-	-
10	-	-	-	-	-	43 90049389 ³²⁾ 90053910 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	66 95005523 ³²⁾ 90053914 ³³⁾	-
16	-	-	-	-	-	53 95005192 ³²⁾ 90053913 ³³⁾	-	53 95005192 ³²⁾ 90053913 ³³⁾	66 95005523 ³²⁾ 90053914 ³³⁾	66 95005523 ³²⁾ 90053914 ³³⁾
25	-	-	-	-	-	-	-	66 95005214 ³²⁾ 90053914 ³³⁾	66 95005523 ³²⁾ 90053914 ³³⁾	66 95005523 ³²⁾ 90053914 ³³⁾

32 MVB - material No. for cable connector, extended and sealed during manufacture

33 OVB - material No. for cable connector, not connected and supplied but not fitted

Table 49: Cable connector: shielded electric extension cable, + or + →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
+ or +										
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	35 95005165 ³²⁾ 90039349 ³³⁾	-	-	-	-	-	-
4	-	-	-	-	-	43 90049390 ³²⁾ 90049400 ³³⁾	-	-	-	-
6	-	-	-	-	-	43 90049390 ³²⁾ 90049400 ³³⁾	-	53 90049392 ³²⁾ 90049402 ³³⁾	66 90033248 ³²⁾ 90055737 ³³⁾	66 90033248 ³²⁾ 90055737 ³³⁾
10	-	-	-	-	-	-	-	66 90039412 ³²⁾ 90053905 ³³⁾	66 90039412 ³²⁾ 90053905 ³³⁾	66 90033248 ³²⁾ 90055737 ³³⁾
16	-	-	-	-	-	-	-	66 90039361 ³²⁾ 90053906 ³³⁾	66 90036538 ³²⁾ 90055738 ³³⁾	66 90036538 ³²⁾ 90055738 ³³⁾
25	-	-	-	-	-	-	-	-	78 90039701 ³²⁾ 90053908 ³³⁾	78 90039701 ³²⁾ 90053908 ³³⁾
35	-	-	-	-	-	-	-	-	-	-
50	-	-	-	-	-	-	-	-	-	-
70	-	-	-	-	-	-	-	-	-	-

Table 50: Cable connector: shielded electric extension cable, →

Motor lead [mm²]	Shielded electric extension cable [mm²]									
	1,5	2,5	4	6	10	16	25	35	50	70
1,5	-	-	-	-	-	-	-	-	-	-
2,5	-	-	-	28 90049385 ³²⁾ 95005106 ³³⁾	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	-	-	-
4	-	-	-	28 90049385 ³²⁾ 95005106 ³³⁾	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	-	-	-
6	-	-	-	28 90049385 ³²⁾ 95005106 ³³⁾	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	-	-
10	-	-	-	-	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	66 95005523 ³²⁾ 90053914 ³³⁾	-
16	-	-	-	-	-	43 90049389 ³²⁾ 90049399 ³³⁾	-	53 90049391 ³²⁾ 90049401 ³³⁾	66 95005370 ³²⁾ 90053914 ³³⁾	66 95005214 ³²⁾ 90053914 ³³⁾
25	-	-	-	-	-	-	-	53 90049391 ³²⁾ 90049401 ³³⁾	66 95005520 ³²⁾ 90053914 ³³⁾	78 95005527 ³²⁾ 05244394 ³³⁾

Motor lead	Shielded electric extension cable [mm ²]									
[mm ²]	1,5	2,5	4	6	10	16	25	35	50	70
										
35	-	-	-	-	-	-	-	53 90049391 ⁽³²⁾ 90049401 ⁽³³⁾	66 95005214 ⁽³²⁾ 90053914 ⁽³³⁾	78 95005527 ⁽³²⁾ 05244395 ⁽³³⁾
50	-	-	-	-	-	-	-	-	66 95005214 ⁽³²⁾ 90053914 ⁽³³⁾	78 95005527 ⁽³²⁾ 05244396 ⁽³³⁾
70	-	-	-	-	-	-	-	-	78 95005527 ⁽³²⁾ 05244396 ⁽³³⁾	78 95005527 ⁽³²⁾ 05244397 ⁽³³⁾

Cable clips

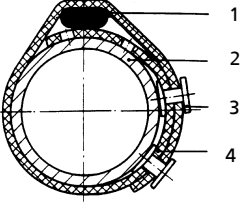
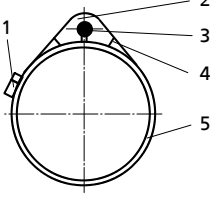
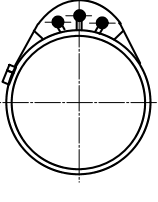
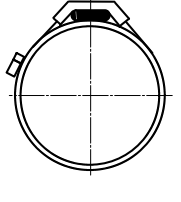
Application

- Securing electric cables at the riser / discharge pipe
- Suitable for use in normal or slightly aggressive water
- Versions for use in more aggressive water (e.g. seawater) on request

Table 51: Selection table

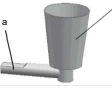
Electric cable			Cable clip size	Cable guard	Mat. No.	Weight	Required strap length L [mm] per cable clip for the following column pipes													
Type	Number x Ø of cores						[kg]	G 11/4 / DN 32	G 11/2 / DN 40	G 2 / DN 50	DN 65	G 3 / DN 80	G 4 / DN 100	G 5 / DN 125	G 6 / DN 150	DN 175	G 8 / DN 200	G 10 / DN 250	G 12 / DN 300	G 14 / DN 350
	[mm]																			
Flat	3 x 1.5 to 3 x 6.0	4 x 1.5 to 4 x 6.0	1	-	01088095	0.04	270	280	320	360	400	450	500	600	-	-	-	-	-	-
	3 x 10 to 3 x 70	4 x 10 to 4 x 70	11	✓	On request	0,01	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900
Round	3 x 1.5 to 3 x 6.0	4 x 1.5 to 4 x 6.0	1	-	01088095	0.04	270	280	320	360	400	450	500	600	-	-	-	-	-	-
	3 x 10	4 x 10	2	✓	On request	0,08	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900
	3 x 16 3 x 25	4 x 16	3a	-	On request	0,19	-	-	350	400	450	550	650	800	950	1100	1300	1500	1700	1900
	3 x 35 3 x 50	4 x 25 4 x 35	3	-	On request	0,19	-	-	-	450	500	600	700	850	1000	1150	1350	1550	1750	1950
	3 x 70	4 x 50	3b	-	On request	0,19	-	-	-	450	500	600	700	850	1000	1150	1350	1550	1750	1950
	3 x 95	4 x 70 4 x 95	4	-	On request	0,55	-	-	-	-	650	750	900	1050	1200	1400	1600	1800	2000	
	3 cables 1 x 50		7	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 70		8	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 95		6	-	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 120		9	✓	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050
	3 cables 1 x 150		9	✓	On request	0,6	-	-	-	-	-	700	800	950	1100	1250	1450	1650	1850	2050

Table 52: Cable clip sizes

Size 1	Sizes 2, 3, 3a, 3b, 4	Sizes 6 to 9	Size 11
 1 = electric cable 2 = riser 3 = plastic stud 4 = rubber strap			
	1 = screw band with tension screw (re-usable) 2 = cable guard 3 = electric cable 4 = metal strap 5 = riser		

Motor refill kit

Table 53: Overview of motor refill kit

	Description	Mat. No.	Weight
			[kg]
	Fill check kit for checking and topping up the motor fill after prolonged storage for UMA 150, UMA-S 150, UMA 200, UMA-S 200, UMA 250, UMA-S 250 motors	90066762	0.25
	Refill kit for checking and topping up the motor fill after prolonged storage for UMA 300 motor	Pipe (a)	0.01
		Filling funnel (b)	0.05

Monitoring and automation

Table 54: Selection table

Description		Mat. No.	Weight [kg/m]
Dry running protection device (supply voltage 220 V / 240 V) for			
	Semi-automatic operation (1 relay, 1 pushbutton + 2 electrodes)	90009553	2
	Fully automatic operation (1 relay + 3 electrodes)	90009554	2
	Plus 1 control cable, blue, 1 x 1.5 mm ² for each electrode; values indicated per metre	01096713	0.1
Immersion electrodes			
	Set of electrodes: 3 electrodes made of stainless steel	40980055	0.3
	Set of electrodes: 1 electrode made of stainless steel	40980056	0.1
Pressure controller , insulating enclosure (membrane type), 1 to 8 bar, with pressure connection G 3/8		01151586	1
Pt100 temperature monitoring equipment for use in submersible motors to protect the windings from overheating. Comprising:			
Pt100 temperature sensor with 10 m cable 4 x 0.5 mm ² for motor:	UMA 150 material variants E, C	01532593	1.1
	UMA 150 material variant D	01532594	1.1
	UMA 200, UMA 250, material variants G, C	90063006	1.1
	UMA 200, UMA 250, material variants D	On request	-
Display / switchgear (supply voltage V = 110/220 V)		90064446	0.45
Control cable, blue, round 4 x 0.5 mm ²		Per metre	01049403
			0.28

Frequency inverter and voltage filter

Table 55: Selection table

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]													
Mat. No.	Mat. No.	Mat. No.	Mat. No.		[kW]	[Hp]	[A]				[A]								
-	01639317	01732881	01732882	UMA-S 150 7/42	4,5	6	8	IP20	48229690	PumpDrive R KSB202 004K00C	10	IP00	01664325	130B2835	33	IP00	01664341	130B2411	13
-	01639317	01732881	01732882	UMA-S 150 7/42	5,5	7	10	IP20	48229692	PumpDrive R KSB202 005K50C	13	IP00	01664325	130B2835	33	IP00	01664341	130B2411	13
-	01639317	01732881	01732882	UMA-S 150 7/42	7,5	10	13	IP20	48229694	PumpDrive R KSB202 007K50C	16	IP00	01664325	130B2835	33	IP00	01664342	130B2412	18
-	01584165	01732855	01732856	UMA-S 150 18/42	9,3	12	18	IP20	48229696	PumpDrive R KSB202 011K00C	24	IP00	01664325	130B2835	33	IP00	01664343	130B2413	28,5
-	01584165	01732855	01732856	UMA-S 150 18/42	11	15	20	IP20	48229696	PumpDrive R KSB202 011K00C	24	IP00	01664325	130B2835	33	IP00	01664343	130B2413	28,5
-	01584165	01732855	01732856	UMA-S 150 18/42	13	17	23	IP20	48229698	PumpDrive R KSB202 015K00C	32	IP00	01664325	130B2835	33	IP00	01664344	130B2281	36
-	01584165	01732855	01732856	UMA-S 150 18/42	15	20	26	IP20	48229698	PumpDrive R KSB202 015K00C	32	IP00	01664325	130B2835	33	IP00	01664344	130B2281	36
-	01584165	01732855	01732856	UMA-S 150 18/42	18,5	25	32	IP20	48229700	PumpDrive R KSB202 018K50C	37,5	IP00	01664326	130B2838	68	IP00	01664345	130B2282	46,5
-	01584166	01732877	01732878	UMA-S 150 37/42	22	29	39	IP20	48229702	PumpDrive R KSB202 022K00C	44	IP00	01664326	130B2838	68	IP00	01664345	130B2282	46,5
-	01584166	01732877	01732878	UMA-S 150 37/42	26	35	46	IP20	48229704	PumpDrive R KSB202 030K00C	61	IP00	01664326	130B2838	68	IP00	01664346	130B3179	86
-	01584166	01732877	01732878	UMA-S 150 37/42	30	40	54	IP20	48229704	PumpDrive R KSB202 030K00C	61	IP00	01664326	130B2838	68	IP00	01664346	130B3179	86
-	01584197	01732879	01732880	UMA-S 150 37/42	37	49	72	IP20	48229708	PumpDrive R KSB202 045K00C	90	IP00	01674691	130B2844	133	IP00	01664346	130B3179	86
-	01639317	01732881	01732882	UMA-S 150 7/42	4,5	6	8	IP55	48229691	PumpDrive R KSB202 004K00W	10	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01639317	01732881	01732882	UMA-S 150 7/42	5,5	7	10	IP55	48229693	PumpDrive R KSB202 005K50W	13	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01639317	01732881	01732882	UMA-S 150 7/42	7,5	10	13	IP55	48229695	PumpDrive R KSB202 007K50W	16	IP54	01664338	130B2837	34	IP54	01664347	192H3025	17
-	01584165	01732855	01732856	UMA-S 150 18/42	9,3	12	18	IP55	48229697	PumpDrive R KSB202 011K00W	24	IP54	01664338	130B2837	34	IP54	01664348	192H3027	24,2
-	01584165	01732855	01732856	UMA-S 150 18/42	11	15	20	IP55	48229697	PumpDrive R KSB202 011K00W	24	IP54	01664338	130B2837	34	IP54	01664348	192H3027	24,2
-	01584165	01732855	01732856	UMA-S 150 18/42	13	17	23	IP55	48229699	PumpDrive R KSB202 015K00W	32	IP54	01664338	130B2837	34	IP54	01664349	192H3028	32
-	01584165	01732855	01732856	UMA-S 150 18/42	15	20	26	IP55	48229699	PumpDrive R KSB202 015K00W	32	IP54	01664338	130B2837	34	IP54	01664349	192H3028	32
-	01584165	01732855	01732856	UMA-S 150 18/42	18,5	25	32	IP55	48229701	PumpDrive R KSB202 018K50W	37,5	IP54	01664339	130B2840	56	IP54	01664350	192H3010	37,5

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter					Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}	
G	E	C	D		[kW]	[Hp]	[A]				[A]				[A]				[A]	
Mat. No.	Mat. No.	Mat. No.	Mat. No.																	[A]
-	01584166	01732877	01732878	UMA-S 150 37/42	22	29	39	IP55	48229703	PumpDrive R KSB202 022K00W	44	IP54	01664339	130B2840	56	IP54	01664351	192H3011	46,2	
-	01584166	01732877	01732878	UMA-S 150 37/42	26	35	46	IP55	48229705	PumpDrive R KSB202 030K00W	61	IP54	01664340	130B2843	83	IP54	01664352	192H3035	61	
-	01584166	01732877	01732878	UMA-S 150 37/42	30	40	54	IP55	48229705	PumpDrive R KSB202 030K00W	61	IP54	01664340	130B2843	83	IP54	01664352	192H3035	61	
-	01584197	01732879	01732880	UMA-S 150 37/42	37	49	72	IP55	48229709	PumpDrive R KSB202 045K00W	90	IP54	01664340	130B2843	83	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	45	60	74	IP20	48229708	PumpDrive R KSB202 045K00C	90	IP00	01674691	130B2844	133	IP00	01664346	130B3179	86	
01738152	-	01738156	01738170	UMA-S 200 75/42	55	74	91	IP20	48229710	PumpDrive R KSB202 055K00C	106	IP00	01674691	130B2844	133	IP00	01733111	130B3182	135	
01738152	-	01738156	01738170	UMA-S 200 75/42	67	90	112	IP20	48229712	PumpDrive R KSB202 075K00C	147	IP00	01674691	130B2844	133	IP00	01733111	130B3182	135	
01738152	-	01738156	01738170	UMA-S 200 75/42	75	100	128	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	75	100	129	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	83	110	143	IP20	48229714	PumpDrive R KSB202 090K00C	177	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	93	125	163	IP20	48229716	PumpDrive R KSB202 110K00C	212	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	100	135	178	IP20	48229716	PumpDrive R KSB202 110K00C	212	IP00	01733056	130B2847	236	IP00	01733112	130B3184	195	
05148497	-	05148498	05148551	UMA-S 200 130/42	110	145	186	IP20	01733781	PumpDrive R KSB202 132K00C	260	IP00	01733056	130B2847	236	IP00	01733113	130B3186	308	
05148497	-	05148498	05148551	UMA-S 200 130/42	130	175	224	IP20	01733783	PumpDrive R KSB202 160K00C	315	IP00	01733056	130B2847	236	IP00	01733113	130B3186	308	
01738152	-	01738156	01738170	UMA-S 200 75/42	45	60	74	IP55	48229709	PumpDrive R KSB202 045K00W	90	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	55	74	91	IP55	48229711	PumpDrive R KSB202 055K00W	106	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	67	90	112	IP55	48229713	PumpDrive R KSB202 075K00W	147	IP54	01733108	130B2846	133	IP54	01674692	192H3037	115	
01738152	-	01738156	01738170	UMA-S 200 75/42	75	100	128	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	75	100	129	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	83	110	143	IP55	48229715	PumpDrive R KSB202 090K00W	177	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	
01738153	-	01738167	01738171	UMA-S 200 100/42	93	125	163	IP55	48229717	PumpDrive R KSB202 110K00W	212	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195	

Motor UMA-S								Frequency inverter PumpDrive R				dv/dt filter				Sine filter			
Material variant				Size	P _N		I _N	Enclosure	Mat. No.	Designation	I _A	Enclosure	Mat. No.	Designation	I _{NFilter}	Enclosure	Mat. No.	Designation	I _{NFilter}
G	E	C	D		[kW]	[Hp]	[A]				[A]				[A]				[A]
Mat. No.	Mat. No.	Mat. No.	Mat. No.																
01738153	-	01738167	01738171	UMA-S 200 100/42	100	135	178	IP55	48229717	PumpDrive R KSB202 110K00W	212	IP23	01733109	130B2848	236	IP23	01733115	130B9027	195
05148497	-	05148498	05148551	UMA-S 200 130/42	110	145	186	IP54	01733782	PumpDrive R KSB202 132K00W	260	IP23	01733109	130B2848	236	IP23	01733116	130B3187	308
05148497	-	05148498	05148551	UMA-S 200 130/42	130	175	224	IP54	01733784	PumpDrive R KSB202 160K00W	315	IP23	01733109	130B2848	236	IP23	01733116	130B3187	308
05147847	-	05147848	05147849	UMA-S 250 200/42	185	246	358	IP20	01839750	PumpDrive R KSB202 200K00C	395	IP00	01733107	130B2849	360	IP00	01845293	130B3188	382
05147847	-	05147848	05147849	UMA-S 250 200/42	200	266	389	IP20	01839789	PumpDrive R KSB202 250K00C	480	IP00	05146486	130B2851	494	IP00	05146604	130B3191	496
05147882	-	05147883	05147901	UMA-S 250 250/42	230	306	447	IP20	01839789	PumpDrive R KSB202 250K00C	480	IP00	05146486	130B2851	494	IP00	05146604	130B3191	496
05147882	-	05147883	05147901	UMA-S 250 250/42	250	333	493	IP20	05117684	PumpDrive R KSB202 315K00C	600	IP00	05146488	130B2853	660	IP00	05146622	130B3193	601
05147847	-	05147848	05147849	UMA-S 250 200/42	185	246	358	IP54	01839790	PumpDrive R KSB202 200K00W	395	IP23	01733110	130B2850	360	IP23	01845295	130B3189	382
05147847	-	05147848	05147849	UMA-S 250 200/42	200	266	389	IP54	01839792	PumpDrive R KSB202 250K00C	480	IP23	05146487	130B2852	494	IP23	05146605	130B3192	496
05147882	-	05147883	05147901	UMA-S 250 250/42	230	306	447	IP54	01839792	PumpDrive R KSB202 250K00C	480	IP23	05146487	130B2852	494	IP23	05146605	130B3192	496
05147882	-	05147883	05147901	UMA-S 250 250/42	250	333	493	IP54	05117687	PumpDrive R KSB202 315K00W	600	IP23	05146597	130B2854	660	IP23	05146623	130B3194	601



KSB SE & Co. KGaA
Johann-Klein-Straße 9 • 67227 Frankenthal (Germany)
Tel. +49 6233 86-0
www.ksb.com